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FINAL REPORT

AI Washing or Credible Disclosure? Patterns and Clusters in Corporate AI Disclosures

By

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I declare that this assignment is my own work, which all sources of reference are acknowledged in full, and it has not been submitted for any other course.

Signature:

How AI disclosure is measured

Table of contents

EXECUTIVE SUMMARY

Corporate AI disclosure expanded rapidly after generative AI became commercially salient: between 2021 and 2025, the share of S&P 500 firms making an AI-related statement rose from 36% to 92%. This thesis asks whether the content of that disclosure distinguishes talk from disclosed substance, and whether that distinction is economically useful.

Three findings organize the evidence. First, firms' AI disclosures fall into recurrent profiles, but only the profile reporting the most concrete activity is consistently economically distinct after sector-year adjustment. Second, specificity and disclosed activity add information beyond disclosure volume and financial fundamentals; promotional tone and the within-disclosure gap between talk and disclosed substance do not. Third, the same firm-year presents a similar activity mix in statutory filings and earnings calls, but calls attach substantially more concrete evidence and promotion. That venue-associated gap widened with the 2023 boom and shows no detectable closure after the SEC's 2024 scrutiny.

The practical implication is not that all promotional AI language is empty. Rather, readers should distinguish volume from specificity, disclosed activity, and cross-venue consistency. The thesis supplies a scalable measurement pipeline and a credibility framework for issuers, investors, and regulators; its claims are associative, not causal.

1. INTRODUCTION

1.1 BUSINESS PROBLEM

The release of ChatGPT in November 2022 turned artificial intelligence from a specialist topic into a mainstream corporate disclosure issue. In this corpus, between 2021 and 2025 the share of S&P 500 firms making at least one AI-related statement in their SEC filings rose from 36% to 92%, and the narrative density of AI disclosure (AI statements per 1,000 words of corporate narrative text) grew nearly seven-fold (from 0.016 to 0.111 per 1,000 words). The single largest jump occurs in the 2024 filing season, when fiscal-year-2023 10-Ks become the first annual reports written with a full year of generative AI behind them: the share of 10-Ks containing any AI statement jumps from 39% to 72% in one season. Independent work also documents a sharp post-ChatGPT increase in 10-K AI disclosure (Anantharaman, Parker, and Rozario 2026, 1–2).

This is, at the same time, an investor problem, an issuer problem, and a regulatory problem:

- **Investors** need to know whether a firm's AI narrative reflects real capability, real deployment, and real economic exposure, or whether it is positioning attached to a hot topic. Studies of filings distinguish actionable or implemented AI from speculative claims and relate that distinction to valuation or later outcomes (Basnet et al. 2025, 1–2; Song et al. 2026, 1–2).
- **Issuers** need to know how much disclosure is enough, and in which venue, without inviting enforcement risk.
- **Regulators**, the SEC prominently among them, have already acted. In February 2024, Chair Gensler warned about "AI washing" (Gensler 2024, 4–5), and the SEC's March 2024 actions

against Delphia and Global Predictions alleged false and misleading statements about AI use under the Advisers Act and Marketing Rule (U.S. Securities and Exchange Commission 2024, 1–2). Those actions make the enforcement risk concrete; they do not establish that every promotional AI statement is unlawful.

The SEC’s April 2025 comment letter to Welltower illustrates why venue comparisons are useful. The staff requested that future filings discuss its data-science platform and the status of AI integration, after noting both the 10-K’s statement that it was integrating generative-AI tools and the earnings-call and press-release descriptions of the platform as “industry-leading” and a source of competitive advantage (U.S. Securities and Exchange Commission 2025, 1). The letter neither finds any statement false nor defines a general cross-venue test; it motivates, rather than validates, the thesis’s cross-venue measurement.

Mention counts, a common measure in finance and disclosure research, tracing back to Loughran and McDonald’s (2011, 36) word lists, cannot answer the semantic questions posed here. A count of how many times “artificial intelligence” appears in a 10-K conflates a risk-factor boilerplate sentence with a specific claim about a named, deployed product generating measured revenue. It cannot:

- separate promotional register from governance register;
- tell whether a claim describes something the firm has done versus something it might do; or
- compare what a firm says in a call against what it commits to in a filing.

Answering the business problem (is the disclosure substantive, and does it matter economically) requires moving past counting mentions and building a measurement pipeline that extracts what is actually claimed: who is the subject, what type of AI, in what tense, in what register, and, one layer deeper, what concrete activity, with what evidence, is behind the claim.

1.2 RESEARCH QUESTIONS

The overarching question this thesis answers is: **can the content of corporate AI disclosure distinguish talk from disclosed substance, and does that distinction carry useful economic information?**

Four linked questions organize the empirical chapters:

- **RQ1: Evolution and composition.** How has corporate AI disclosure evolved, and what activities do firms report?
- **RQ2: Heterogeneity.** What recurrent disclosure profiles distinguish firms, and how do those profiles relate to firm economics?
- **RQ3: Economic information.** Does semantic content and disclosed activity add information beyond disclosure volume and financial fundamentals?
- **RQ4: Credibility and decoupling.** Where do gaps emerge between disclosure and disclosed substance, and between statutory filings and earnings calls?

1.3 LITERATURE ON CORPORATE AI DISCLOSURE AND AI WASHING

1.3.1 THE EMERGING EMPIRICAL FRONTIER

The explosion of commercial interest in artificial intelligence following the release of ChatGPT triggered a rapid empirical literature. Studies examine AI language in annual filings (Basnet et al. 2025, 1; Song et al. 2026, 1–2; Anantharaman, Parker, and Rozario 2026, 1–2) and firm-level generative-AI exposure constructed from occupational tasks and employment shares (Eisfeldt,

Schubert, and Zhang 2023, 9). This work builds on the broader textual-analysis tradition in finance (Li 2008, 1–3; Loughran and McDonald 2011, 35–36; Gentzkow, Kelly, and Taddy 2019, 1–2).

Simultaneously, a regulatory and market concern has arisen: “AI washing,” the use of AI claims that are false, misleading, or unsupported by observable implementation. Gensler used the term in his warning and the SEC used it in the Delphia and Global Predictions release (Gensler 2024, 4–5; U.S. Securities and Exchange Commission 2024, 1–2). Borrowing conceptual scaffolding from greenwashing and corporate environmental decoupling (Delmas and Burbano 2011, 64–68; Lyon and Montgomery 2015, 1–3; Marquis, Toffel, and Zhou 2016, 483–85), recent work contrasts speculative AI narratives with observable implementation or actionable disclosures (Song et al. 2026, 1–2; Basnet et al. 2025, 1–2).

1.3.2 FOUR GAPS IN THE CURRENT LITERATURE

Despite rapid progress, the existing literature on corporate AI disclosure and AI washing has four gaps:

1. **Keyword retrieval alone treats mentions too coarsely.** Some AI-disclosure studies begin with keyword retrieval from annual reports or 10-Ks (Anantharaman, Parker, and Rozario 2026, 1–2). Others improve on it by classifying mentions as actionable, speculative, or irrelevant (Basnet et al. 2025, 1; Song et al. 2026, 1–2). Keyword retrieval by itself cannot tell an empty risk-factor sentence (“new AI tools developed by competitors could disrupt our market position”) apart from a specific claim about a named, deployed product with a measured result (“we deployed an internal predictive model that cut supply-chain logistics costs by 14%”).
2. **The cited studies do not provide the same activity inventory.** Basnet et al. classify filing narratives into actionable, speculative, and irrelevant mentions; Song et al. distinguish planned from implemented AI using text and patents; and Eisfeldt et al. construct firm-level generative-AI exposure from task and occupational-employment data (Basnet et al. 2025, 1; Song et al. 2026, 1–2; Eisfeldt, Schubert, and Zhang 2023, 9). These designs answer different questions from a structured inventory of the business function, technology, sourcing, maturity, and evidence attached to each disclosed activity.
3. **The cited AI studies use annual filings rather than a matched call–filing design.** Earnings calls are a voluntary disclosure channel (Matsumoto, Pronk, and Roelofsen 2011, 1383–84), and research on calls documents managers’ incentives and discretion in that channel (Frankel, Johnson, and Skinner 1999, 133–34). Comparing the same firm-period in calls and filings therefore adds a venue dimension absent from the annual-filing designs cited above. The Welltower letter is a concrete example in which the SEC staff referenced both types of communication (U.S. Securities and Exchange Commission 2025, 1).
4. **No complete external census validates firm-level AI adoption.** Greenwashing research can compare claims with environmental performance indicators (Delmas and Burbano 2011, 68–69; Marquis, Toffel, and Zhou 2016, 486–87). AI research can use partial observables such as patents (Song et al. 2026, 1–2), but these do not constitute a complete census of deployment across business functions. The thesis therefore measures disclosed activity and its grounding, not actual adoption.

1.3.3 WHAT THIS THESIS DOES ABOUT IT

This thesis addresses these four gaps in four measurement and design steps:

1. **From keyword counts to structured claims.** Instead of counting words or computing hard-to-interpret text embeddings, the pipeline filters millions of paragraphs with an interpretable

classifier, then has an LLM judge extract one structured claim per statement: who is speaking, what type of AI, in what tense (realized, planned, expected, hypothetical), about what business domain, using which concept from a fixed vocabulary, how specific, how promotional, and with what sentence-level evidence.

2. **An inventory of what firms say they do, not just a number for a regression.** Instead of reducing disclosure text to a single score, this thesis builds a descriptive inventory of 27,566 concrete activity claims (action, object, and business function) across 471 S&P 500 firms, recording the stage, audience, evidence, and technology source (in-house vs. third-party) behind each one. This is an inventory of what firms *say* they do, not an audited record of what they operationally do.
3. **Comparing the same firm across venues.** Firm $\times$ fiscal-year fixed effects compare the call and the filing while holding the firm and period fixed. This design measures a venue-associated difference; it does not isolate the causal effect of legal liability, because audience, format, and topic selection also vary with venue.
4. **A single AI-washing index, covering every firm-year that discloses AI.** Rather than an arbitrary tercile cut or a claim-by-claim ratio restricted to a highly promotional subsample, the thesis defines AI washing as the percentile-rank gap between disclosure intensity and concrete, grounded disclosed activity. The measure is defined for every firm-year that mentions AI and is examined alongside the cross-venue comparison as a distinct form of decoupling.

1.4 THEORETICAL FOUNDATIONS

Three ideas from economics, accounting, and organizational sociology motivate the empirical designs that follow.

Signalling and cheap talk. Spence's (1973, 363–64) signalling model makes signal cost central to credibility; voluntary-disclosure theory likewise models the costs of releasing information (Dye 1985, 123–24). Crawford and Sobel (1982, 1431–32) analyze strategic information transmission when sender and receiver interests diverge. These models do not prove that a named model, quantified gain, or commercial partner is costly or true. They motivate the empirical distinction: such details are potentially more verifiable than generic promotional language, while tone itself can be associated with litigation risk (Rogers, Van Buskirk, and Zechman 2011, 2155–56).

Institutional pressure to look current. Organizational sociology (Meyer and Rowan 1977, 340–41; DiMaggio and Powell 1983, 147–48; Suchman 1995, 571–74) argues that under technological uncertainty, firms adopt the language and symbols expected of them to stay legitimate, whether or not that language reflects any change in what they actually do. This is a plausible reason why AI disclosure spread from 36% to 92% of the S&P 500 in four years, including in sectors where AI has limited operational relevance.

Talk that doesn't match conduct. Meyer and Rowan (1977, 340–41) distinguish formal, ceremonial structure from ongoing activity; Westphal and Zajac (1994, 367–69) and Crilly, Zollo, and Hansen (2012, 1429–31) examine forms of policy–practice decoupling. Crilly et al. contrast intentional decoupling (“faking it”) with implementation friction (“muddling through”). In corporate disclosure, related concerns appear in work on impression management and tone (Merkl-Davies and Brennan 2007, 116–18; Huang, Teoh, and Zhang 2014, 1083–86). These theories motivate the thesis's disclosed-activity measures; they do not turn a disclosure–activity gap into proof of actual nonimplementation.

Four terms used throughout (disclosure intensity, disclosed behaviour, activity grounding, AI washing):

- **AI disclosure intensity:** how much a firm says about AI, measured from the semantic-frame pass as AI frames per 1,000 words of a firm's own filings that year. It is distinct from *promotional posture*, the share of a firm's AI statements written in promotional, strategic, or aspirational language.
- **Disclosed behaviour (disclosed activity volume):** measured from a separate LLM extraction pass over concrete, action-level claims (Chapter 2's disclosed-activity inventory), kept deliberately apart from the frame-level measures used to build disclosure intensity, so the two remain distinct instruments rather than restatements of the same content, though both still read the same underlying text, so any relationship between them is descriptive, not independently confirmed.
- **Activity grounding:** a firm-year score for how concretely disclosed activities are specified (named business function, deployed/scaled stage, named product, quantified outcome, named provider).
- **AI washing (the index):** the percentile-rank gap between AI disclosure intensity and disclosed substance — disclosed activity volume weighted by activity grounding — defined only for firm-years that disclose AI at all. It is one within-disclosure form of decoupling; the related cross-venue gap compares calls and filings.

Propositions mapped to empirical research questions.

- *RQ1 (Evolution and composition):* Documents the growth and changing composition of AI disclosure, then inventories the activities firms report.
- *RQ2 (Heterogeneity):* Tests whether disclosure posture organizes into stable archetypes and whether those profiles correspond to economically distinct firms after sector-year adjustment.
- *RQ3 (Economic information):* Tests whether specificity and concrete activities carry incremental economic information beyond disclosure volume and promotional tone.
- *RQ4 (Credibility and decoupling):* Measures the within-disclosure percentile gap and the within-firm, within-year call-versus-filing gap; the SEC exercise is a secondary identification check within this question.

1.5 CONTRIBUTION

The thesis makes three contributions.

1. Measurement. A scalable five-layer pipeline separates AI disclosure *volume*, *semantic content*, and *disclosed activity* across SEC documents and earnings-call transcripts.

2. Empirical design. The analysis separates the evolution and composition of disclosure, recurrent disclosure profiles, incremental economic information, and credibility gaps rather than treating them as a single mention-count exercise.

3. Credibility framework. A within-disclosure disclosure–substance gap and a within-firm, cross-venue comparison distinguish two forms of decoupling without treating either as direct evidence of actual operational conduct.

Managerial contribution. For issuers, the thesis converts the measurement into a concrete checklist for credible disclosure (named activity, clear stage, quantified outcome, provider transparency, venue consistency) and quantifies a venue gap regulators are already scrutinizing. For investors, it identifies which textual dimensions carry incremental information (specificity and disclosed activity,

not promotional tone) and shows that cross-venue consistency between the call and the filing is itself a usable credibility signal.

1.6 THESIS ROADMAP

- **Chapter 2** documents the corpus, the measurement pipeline, and its human validation.
- **Chapter 3** asks what firms disclose about AI: how disclosure grew and changed in composition 2021–2025, what activities firms describe doing, and whose technology they say it is.
- **Chapter 4** asks how firms differ: six conceptually defined dimensions of AI disclosure posture, the archetypes built on them, and their economic characterization.
- **Chapter 5** asks whether semantic content and disclosed activity, beyond disclosure volume, carry incremental economic information.
- **Chapter 6** examines credibility and decoupling: the within-disclosure washing index, the cross-venue gap, and whether the SEC’s 2024 scrutiny produced a detectable break.
- **Chapter 7** draws together the issuer, investor, and theoretical implications without repeating the empirical results.
- **Chapter 8** concludes.

Table 1: Theory to empirical test.

Theory	What is measured	Hypothesis / RQ	Empirical test	Chapter
Institutional diffusion / legitimacy (DiMaggio and Powell 1983, 147–48; Suchman 1995, 571–74)	Share of firms disclosing, “no AI” segment	RQ1	Evolution 2021–2025, cohort check	3
Recurrent disclosure archetypes (Hennig 2007, 1–2)	Six disclosure-posture dimensions, then K-means on those dimensions	RQ2	K-means, bootstrap Jaccard stability, PCA robustness check	4
Economic coherence of archetypes	Segment × sector-year residual profiles	RQ2	Sector-year residualized regressions	4
Signalling / cheap talk (Spence 1973, 363–64; Crawford and Sobel 1982, 1431–32)	Specificity and disclosed activity vs. promotional rhetoric	RQ3	Nested regressions, ΔR^2 , Wald, permutation	5
Decoupling within and across venues (Meyer and Rowan 1977, 340–41; Crilly, Zollo, and Hansen 2012, 1429–31)	Disclosure-substance gap; call – filing gap	RQ4	Percentile ranks; firm × fiscal-year fixed effects	6
Enforcement credibility (Gensler 2024, 4–5; U.S. Securities and Exchange Commission 2024, 1–2; Callaway, Goodman-Bacon, and Sant’Anna 2024, 1–2)	Pre/post SEC scrutiny	RQ4	Event study with continuous exposure, DiD	6

2. DATA AND MEASUREMENT

2.1 CORPUS AND SOURCES

The U.S. corpus covers 10-K, 10-Q, 20-F, DEF 14A, and 8-K filings for the S&P 500 universe plus earnings-call transcripts, from 2021 through a partial 2026. The document counts and deduplication

totals below are calculated directly from the current processed corpus at render time; the U.S. instrument carries embeddings, prefilter scores, and semantic frames, and is the basis for every result in this thesis.

The current processed corpus contains **58,210 documents** and **9,746,218 paragraph instances**. The length-and-content filter retains 8,900,590 scorable instances, which deduplicate to 5,107,370 unique texts.

Table 2: Current processed U.S. corpus by form.

Form	Documents
10-K	2,898
10-Q	8,191
20-F	32
DEF 14A	2,880
8-K	35,880
Earnings call	8,329
Total	58,210

The measurement pipeline moves through four stages: the raw corpus, a machine-learning prefilter, semantic-frame extraction, and disclosed-activity extraction. The unit of analysis changes at each stage (document, deduplicated text, individual claim, concrete activity) because each stage needs a different unit to do its job. Conceptually, this multi-stage inference architecture instantiates the classical Knowledge Discovery in Databases (KDD) process (Fayyad, Piatetsky-Shapiro, and Smyth 1996, 40–41)—progressing through selection, preprocessing, transformation, data mining, and pattern evaluation—and aligns with the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework (Shearer 2000, 15–19), mapping regulatory and corporate understanding into disciplined, reproducible data engineering and model evaluation stages. [Figure 1](#) illustrates this end-to-end data acquisition, inference, and multi-source fusion architecture.

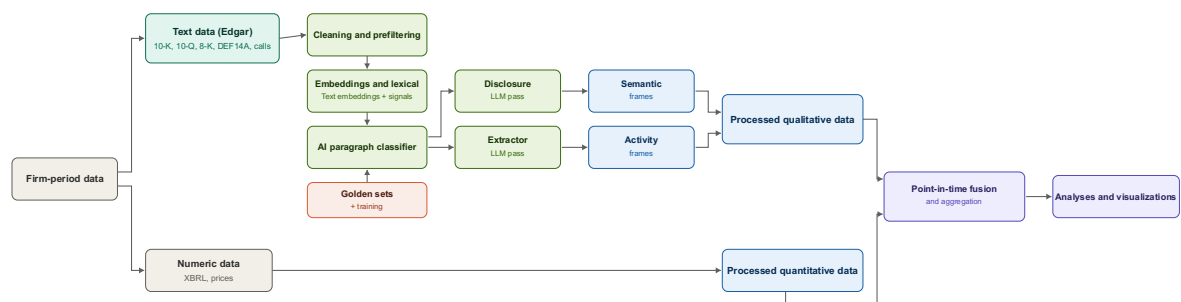


Figure 1: End-to-end data acquisition, inference, and multi-source fusion pipeline.

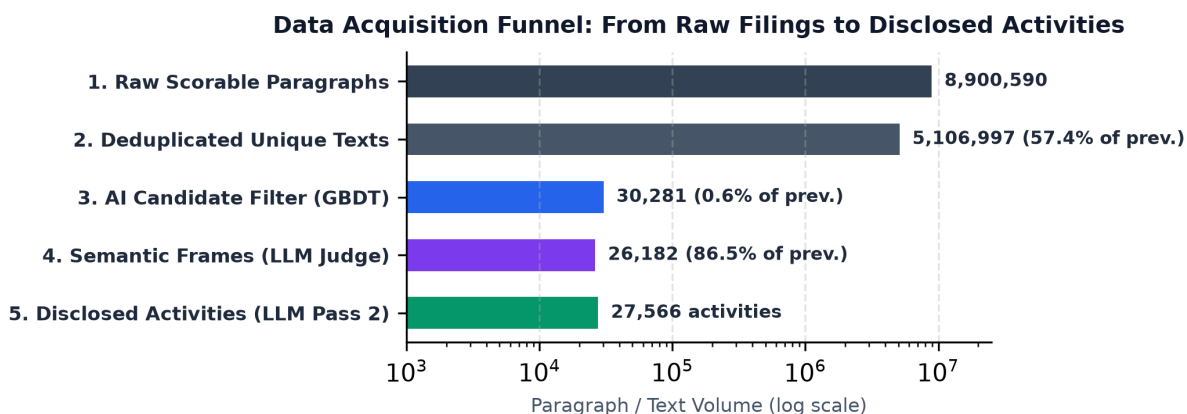


Figure 2: The Data Acquisition Funnel: progression of corpus volume across analytical screening stages (log scale).

Table 3: Compact summary of the inference funnel.

Stage	Unit of Observation	Operational Filter	Retained Volume	Pipeline Share
1. Raw Corpus	Paragraph instances	Universe coverage (S&P 500, 2021-2026)	8,900,590	100.00%
2. Deduplication	Unique scorable texts	BLAKE2b hash (> 3 char, alphanumeric)	5,106,997	57.38%
3. Machine Learning Gate	Candidate paragraphs	Supervised GBDT classifier (tau=0.17)	30,281	0.59%
4. Semantic Frames	Texts with ≥ 1 frame	Zero-shot LLM judge (qwen3.7-flash)	26,182	0.51%
5. Disclosed Activities	Behavioral action triples	Firm-conditioned LLM extraction (pass 2)	27,566	0.54%

Notes: High-level overview of the five-stage inference pipeline. Granular form-by-form decompositions (10-K, 10-Q, DEF 14A, 8-K, and Earnings Calls) and intermediate model acceptance diagnostics are reported in [Table 23](#) (Appendix C).

2.2 UNITS OF ANALYSIS

The pipeline moves across seven nested units, each used where it is the natural denominator:

- the unique text (a deduplicated paragraph, the unit of scoring and classification);
- the semantic frame (one claim extracted from a text; a text can yield zero, one, or several frames);
- the disclosed activity (one action-object-function triple extracted from a behavioural frame);
- the document (the unit of intensity, since a text's frames are joined back to every document instance in which it appears);
- the firm-year (the unit of the economic analysis);
- the pooled firm (the unit of the cross-sectional segmentation and washing score); and
- the firm $\times$ fiscal-year $\times$ venue cell (the unit of the cross-venue comparison).

A firm-quarter unit is used only for the SEC and DeepSeek event studies, where calendar-quarter alignment is required to define a clean pre/post cut.

2.3 CLEANING, PREFILTERING, AND AI PARAGRAPH CLASSIFIER

Before any LLM reads a paragraph, the text processing stack cleans, deduplicates, and classifies every candidate paragraph. Every scorable paragraph is hashed on its normalized text, which collapses 8.73 million paragraph instances into 4.98 million unique texts. Everything downstream runs once per unique hash rather than once per document, and results are joined back afterward to every filing that repeats that paragraph. This keeps recycled boilerplate and legal disclaimers from being scored repeatedly across documents.

Text embeddings and lexical signals are computed for each unique paragraph. A gradient-boosted classifier evaluates 39 signals across four groups:

- keywords (a hand-built list of AI terms and roughly 60 named entities);
- meaning (cosine similarity to known AI language using dense sentence embeddings);
- shape (length, capitalization, position in the document); and
- sentence-level cues.

It is trained on a 9,900-text golden set labeled by two independent LLM judges ([Table 6](#)) and scores 0.96 on the standard F1 metric for 10-K filings, meaning it rarely misses a real AI mention and rarely flags something irrelevant, on the document type it was tuned for. Precision is recovered downstream in the LLM passes, not here.

On a held-out sample never used in training, precision (the share of flagged texts that are actually about AI) stays at 0.98 for 10-K/10-Q, drops to 0.73 for proxy statements and 8-Ks, and drops further to 0.59 for earnings calls. Recall (the share of true AI mentions the filter catches) stays above 0.92 everywhere, so the filter almost never misses something real; it just lets through more noise on some document types than others.

In total, it flags 30,281 texts as AI-relevant, 0.6% of everything scored. Every downstream number in this thesis counts a document with zero flagged texts as a documented zero, not as a missing observation: a firm that never mentions AI is data, not a gap in the sample.

2.4 DISCLOSURE LLM PASS: SEMANTIC FRAMES

Every text the prefilter flags is then read once by an LLM. Prior computational social-science work finds that LLM labels can be comparable to human labels in specific, evaluated tasks (Gilardi, Alizadeh, and Kubli 2023, 1–2); that evidence does not remove the need for the held-out evaluation reported here. The pass pulls out zero, one, or several separate claims about AI, a “frame.” Each frame records eight things about the claim:

- who it’s about (the firm itself, a customer, a competitor, a supplier);
- what kind of AI (generative vs. predictive/ML: generative went from roughly half of mentions in 2021 to 93% by 2026);
- its tense (already happening, planned, expected, or hypothetical: 72% describe something already happening);
- its business domain;
- which concept from a fixed list it matches (adoption, capability, risk, or governance);
- how specific it is;
- whether it reads as promotional; and

- what evidence backs it.

This step also acts as a second, more careful filter: 13% of what the cheap prefilter flagged turns out, on a closer read, to have zero real AI claims and gets dropped. In total, 30,281 texts are read this way, yielding 43,366 distinct claims.

Table 4: Semantic frame schema: the eight fields recorded per claim, with observed value distributions.

Field	Description	Observed values
subject	Who the claim is about	firm (84%); competitors_or_industry (8%); customers (7%); suppliers_or_partners (1%)
ai_type	Generative vs. predictive/ML	unspecified (57%); generative (37%); predictive_ml (6%)
temporal	Tense: realized, planned, expected, hypothetical	realized (71%); planned (14%); expected (10%); hypothetical (5%)
domain	Internal, customer-facing, or unspecified	customer_facing (44%); internal (43%); unspecified (13%)
concepts	Fixed vocabulary of adoption/capability/risk/governance concepts (multi-label)	reported descriptively by frame type
specificity	5 sub-flags: process, product, vendor, metric, date/timeline named	combined into a firm-level specificity measure
rhetoric	Promotional or strategic-importance framing (boolean)	14.9% promotional (corpus mean)
evidence	Sentence indices supporting the claim	always non-empty by construction

2.5 EXTRACTOR LLM PASS: ACTIVITY FRAMES

Whenever a claim describes the firm itself doing something concrete (deploying, investing, hiring, measuring a result), it goes through a second pass. This time the model is told which company it's reading, so it can tell "our own product" apart from "a tool we bought." For each such claim it records:

- the action (deploy, develop, partner, invest in infrastructure, hire, pilot, scale, measure a result, govern, restrict, and a few others);
- what the action targets (in free text, later grouped into families) and who benefits from it (employees, customers, internal process, partners);
- how mature it is (exploring, piloting, deployed, scaled);
- whose technology it is (the firm's own, a third party's, co-developed, acquired, open-source);
- who else gets named (a partner, a vendor, a competitor, a customer); and
- how solid the evidence is (a named product, a number, a named vendor, or just generic language).

Every one of these records is tied to an actual sentence; nothing is inferred without a supporting quote. This pass produces 27,566 distinct activities across 471 of the 510 firms.

Table 5: Disclosed activity schema: the fields recorded per activity, with observed value distributions.

Field	Description	Observed values
action	What the firm did	deploy (63%); invest_infrastructure (8%); scale (7%); pilot_or_explore (4%); ...
stage	Operational maturity	deployed (58%); scaled (14%); piloting (10%); exploring (9%); ...

Field	Description	Observed values
ai_source	Whose technology it is	own (73%); unspecified (16%); third_party (8%); acquired (1%); ...
target	Who benefits	customers (48%); internal_process (25%); unspecified (14%); employees (9%); ...
evidence_strength	How solid the evidence is	generic (53%); named_product_or_process (29%); metric (13%); vendor (4%)
object	Free text, later grouped into 21 object families	reported as an action × object typology
named_entities	Own brand, external provider/model, partner, acquired company, customer, distribution channel, competitor	17,193 entity mentions total

2.6 POINT-IN-TIME FUSION AND AGGREGATION

The measurement pipeline distinguishes between two complementary concepts of disclosure volume:

1. **Diffusion:** the share of filings (or of firms) with at least one AI statement. Bounded between 0 and 1, and unaffected by how many separate claims a document makes.
2. **Cumulative frame density:** the total number of AI statements per 1,000 words of scorable narrative text: how much AI content there is, not just whether it's present. Every "per 1,000" intensity figure in this thesis, including in the clustering, regression, and event-study analyses, uses words as the denominator rather than paragraphs, so a single unit of measurement holds throughout.

The two move together closely (a firm with more AI content is almost always also a firm that discloses at all), so it makes little difference which one a ranking or a regression uses. On average, a paragraph that discusses AI makes 1.44 separate statements, rising from 1.43 in 2021 to 1.49 in 2025 as filings got denser.

To go from a single document to a firm-year, each document's rate is combined with the others using the median, not an average weighted by length. The reason is practical: a firm-year can include one 10-K and several 10-Qs and 8-Ks of very different sizes, and the median stops one unusually long or short document from dominating the number. In practice this choice barely matters, because the 10-Q almost never contains an AI statement, so a firm-year's AI content usually comes from a single filing (its 10-K) anyway, and the median gives the same answer as a length-weighted average to the ninth decimal place across the analytical firm-year panel (checked directly, not assumed).

For concepts describing behaviour (deployment, risk, governance, etc.), the default measure is a log-transformed count rather than a share of a firm's total AI statements: the count version turned out to be much more reliable when the same firm was checked against itself twice (a reliability score of 0.845 vs. 0.528 for shares).

Firm-level rates are also pulled slightly toward the corpus average when a firm has very few AI statements to base a rate on, so a firm with two mentions doesn't get treated as confidently as one with two hundred; a firm with zero mentions gets no rate at all and goes into its own "no AI" category instead.

Activity grounding is simpler: the average of five yes/no questions per activity:

- is the function named;
- is it deployed or scaled;
- is the product named;

- is there a number attached; and
- is the provider named.

2.7 GOLDEN SETS AND MODEL TRAINING

Every automated stage in the measurement pipeline is benchmarked against held-out ground truth and inter-annotator evaluation datasets. [Table 6](#) details the evaluation datasets, their sample design, and empirical performance metrics.

Table 6: Labeled evaluation datasets, validation benchmarks, and inter-judge agreement.

Benchmark Dataset	Forms	Sample Size	Annotated By	Metric / Empirical Performance	Status
Golden Set (Training)	10-K, 10-Q	9,900 texts	Qwen 3.7 / Gemini 3.8	F1 = 0.96 (Prec. 0.98, Rec. 0.94)	Verified
Inter-Judge Agreement	10-K, 10-Q	6,038 pairs	Gemini 3.8 vs. Qwen 3.7	Cohen's κ = 0.81–0.87 across dimensions	Verified
Out-of-Domain Proxy	DEF 14A, 8-K	1,500 texts	Qwen 3.7 Flash	Prec. 0.73, Rec. 0.92, F1 = 0.82 (14A), 0.65 (8-K)	Verified
Out-of-Domain Calls	Transcripts	800 texts	Qwen 3.7 Flash	Prec. 0.59, Rec. 0.98, F1 = 0.73 (Strong Prec. 0.85)	Verified
Lexical Anchors & Gate	All forms	27 anchors	Expert curated	39 signals, BGE-M3 cosine anchors	Verified
Activity Extraction	Mixed sample	30 paragraphs	Iterative manual inspection	Entity role & source prompt engineering	Verified
Human Ground Truth	All forms	720 texts	Human coders (ui-validator/)	Human-to-LLM agreement (434 frames, 267 activities)	In progress

Notes: All operational labels supporting the empirical chapters derive from zero-shot LLM extraction. The measured inter-judge agreement (κ = 0.81–0.87) establishes strong reproducibility across distinct foundation model architectures (Google Gemini vs. Alibaba Qwen).

2.8 PROCESSED QUANTITATIVE DATA: FINANCIAL AND MARKET VARIABLES

Firm-year financial and market variables are drawn from XBRL and price data, aligned by actual fiscal period end rather than filing date, with documented fallback chains for concept coverage (capex 87%, SG&A 79%, net margin 98%, revenue 98%) and a [340, 380]-day gap rule for validating year-over-year changes. The variable set used throughout Chapters 3–5 comprises market capitalization, R&D-to-sales, gross margin, operating margin, asset turnover, price-to-sales, next-fiscal-year revenue growth, market beta, and idiosyncratic volatility (the annualized standard deviation of one-factor market-model residuals over the 252 trading days preceding the filing). R&D-to-sales has only 45% XBRL coverage and is used as an outcome, not a general control: it identifies firms that separately report comparable current R&D rather than treating non-reporting as zero. Appendix D gives the beta specification, its estimation-window diagnostics, the treatment of delisted firms, and the R&D coverage assessment. All financial ratios are winsorized at the 1st and 99th percentiles.

3. WHAT FIRMS DISCLOSE ABOUT AI

3.1 EVOLUTION 2021–2025

The expansion of AI disclosure can be tracked on two complementary margins: **diffusion** (the share of S&P 500 firms making at least one AI statement in their SEC filings) and **cumulative frame density** (AI statements per 1,000 words of scorable narrative text). Diffusion rises from 35.7% in 2021 to 92.1% in 2025. Concurrently, cumulative frame density rises from 0.016 to 0.111 frames per 1,000 words (a 6.9-fold increase).

The ratio of frames per AI-mentioning paragraph expands from 1.43 in 2021 to 1.49 in 2025, showing that as AI disclosure diffused, disclosing paragraphs also became semantically denser, frequently addressing risk, governance, and operational rollout in the same passage. The single largest discontinuity occurs in the 2024 filing season, when fiscal-2023 10-Ks (the first written with a full year of generative AI exposure) push the share of 10-Ks with any AI content from 39% to 72%. A balanced cohort of the 475 firms present in both 2021 and 2026 reproduces the identical trajectory of rising diffusion, confirming that market-wide diffusion rather than sample attrition drives the trend.

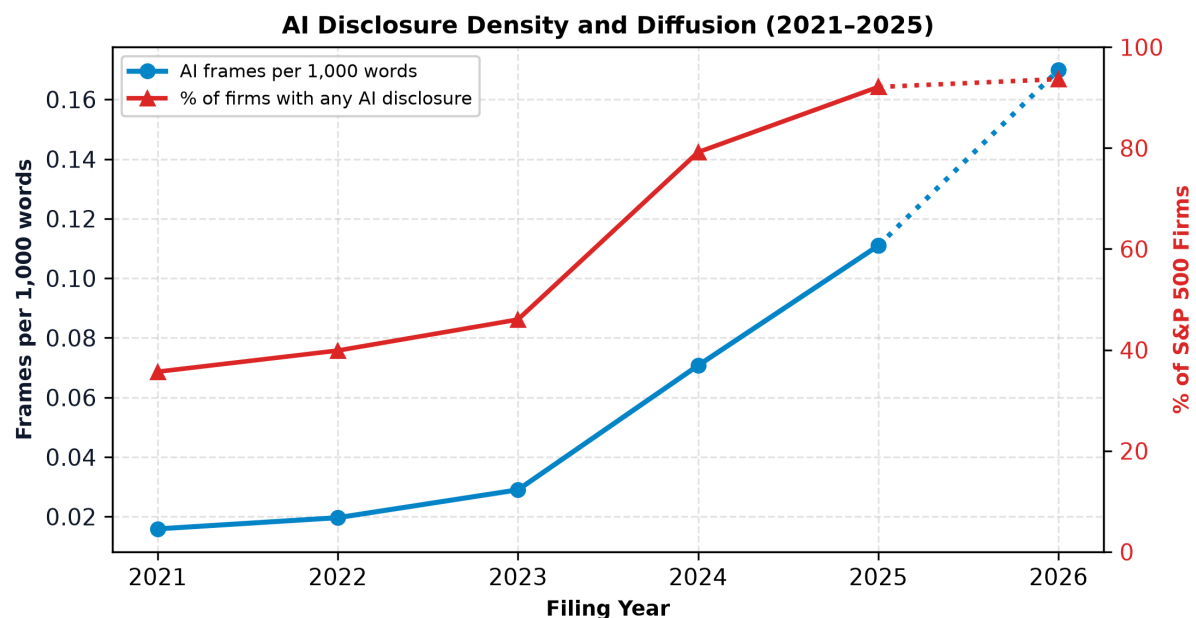


Figure 3: Evolution of AI disclosure density (AI statements per 1,000 words of scorable narrative text) alongside the share of disclosing S&P 500 firms, 2021–2025.

The composition of that growth matters more than its scale: risk grows roughly five times faster than promotion, and governance three times faster:

Table 7: Frame-type growth and composition, 2021-2025.

Frame type	Growth, 2021-2025	Share of firm's AI frames, 2021	Share, 2025
Risk	20.3x	11.4%	33.2%
Governance	11.7x	7.2%	12.1%
Realized-outcome	6.0x	80.8%	70.0%
Promotional	4.2x	13.7%	8.2%
Deployment	4.2x	51.1%	30.7%

Notes: Shares are of a firm's own AI frames and need not sum to 100%, since a single frame can carry more than one concept (e.g., a realized, risk-related statement).

What firms added to their filings between 2023 and 2025 was, above all, an AI risk factor and a governance line; this composition leaves open whether product-oriented communication is concentrated in another venue.

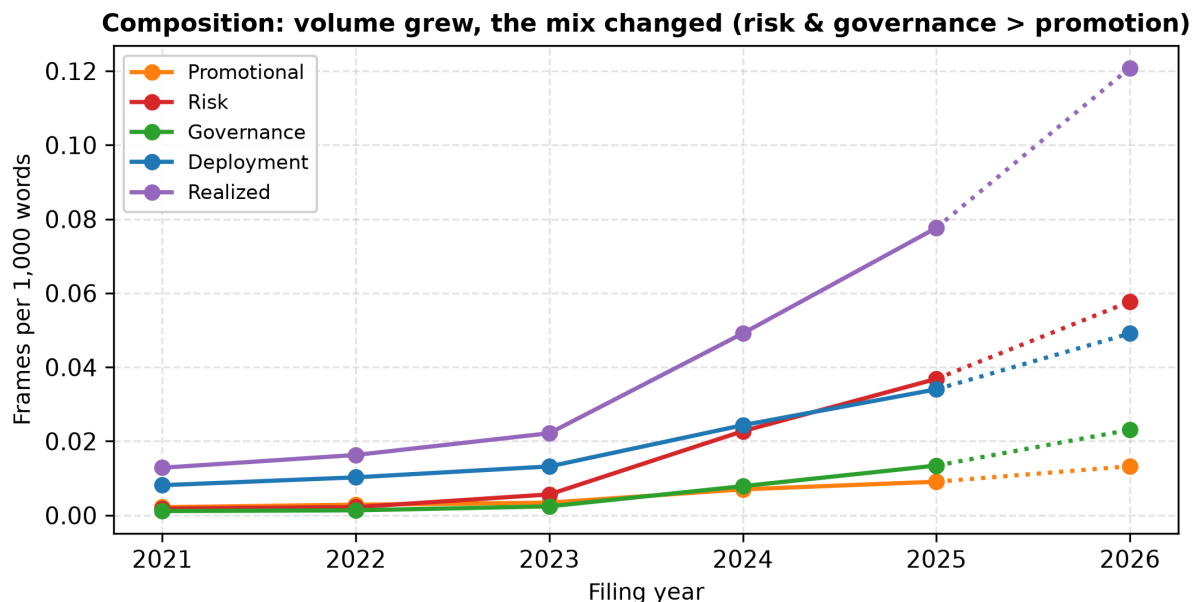


Figure 4: Evolution of semantic frame composition in corporate filings (2021–2025), showing the rapid rise of risk and governance disclosures relative to promotional claims.

3.2 WHAT FIRMS SAY ABOUT AI

Before any clustering, the raw distribution of frame concepts already separates quantity from composition:

- Realized framing dominates temporal claims (72.1%) over planned (12.5%), expected (8.2%), and hypothetical (7.2%) claims.
- The firm itself is the subject of 85.3% of statements, versus 9.1% for competitors and industry, 4.9% for customers, and 0.6% for suppliers and partners.
- Internal use narrowly exceeds customer-facing use in filings (46.6% vs. 37.7%), while the ratio inverts in earnings calls (35.1% internal vs. 55.6% customer-facing): the filing describes process, the call describes product.
- Cybersecurity, regulatory/legal exposure, and competitive disruption dominate the risk vocabulary; generative AI overtakes predictive/ML mentions between 2022 and 2023 and reaches roughly 93% of type-declared mentions by 2026.

These raw distributions motivate, but do not by themselves establish, the archetype structure developed next.

What firms say about AI, before any segmentation

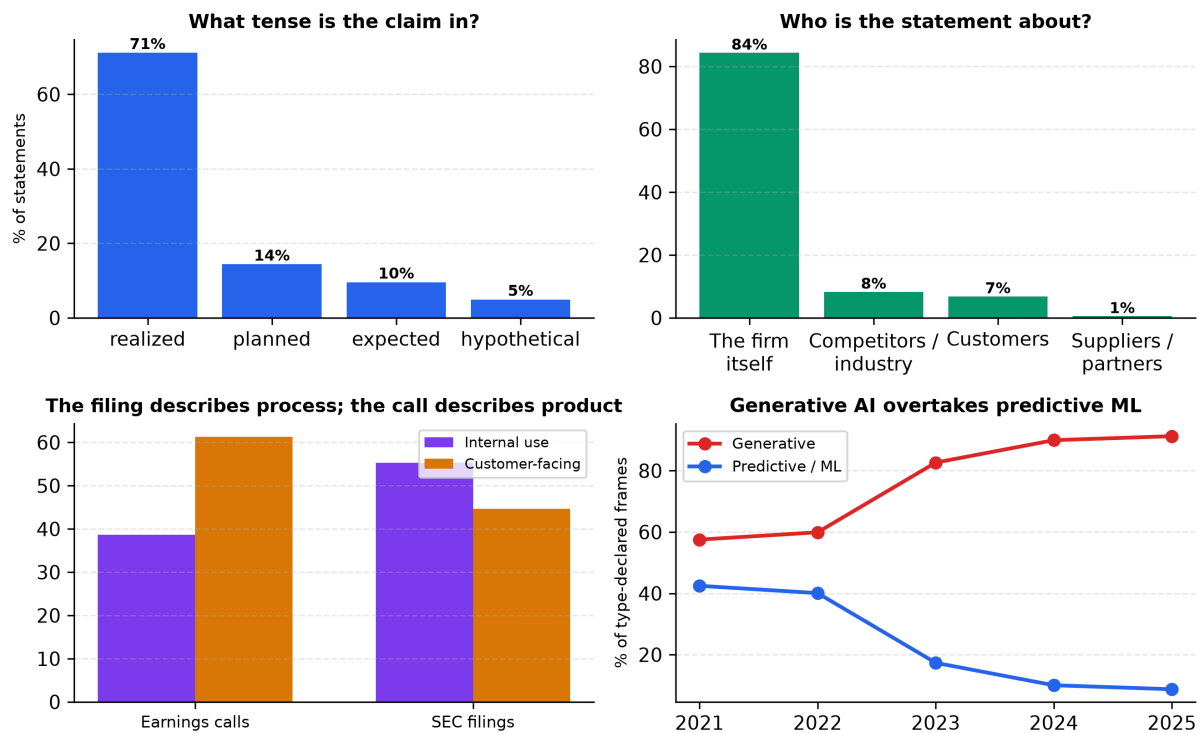


Figure 5: The raw semantic landscape before any clustering: temporal framing, who is speaking, internal vs. customer-facing use by venue, and the generative/predictive shift over time.

3.3 DESCRIPTIVE INVENTORY OF DISCLOSED CORPORATE AI ACTIVITIES

To see what public corporations literally disclose doing with artificial intelligence, rather than reducing that to a single number, the second-stage extraction pipeline recovers 27,566 unique disclosed activities across 471 S&P 500 firms (92.4% of firms with filings). Every activity represents an extracted (action, object, function) triple grounded in a verified sentence of textual evidence, annotated with its operational maturity stage, target audience, technology origin, and named entities.

3.3.1 AGGREGATE BEHAVIORAL LANDSCAPE, 2021–2025

Six dimensions describe what a firm is doing, how mature it is, for whom, whose technology it is, how well evidenced the claim is, and which business function it serves, and each of the six moved between 2021 and 2025, in the same direction: toward less maturity and less concreteness per activity, even as the number of activities disclosed each year grew several-fold.

- The deployed share of activities falls from 65% to 59%, while piloting and exploring combined rise from 16% to 18% (peaking at 25% in 2023).
- The own-technology share drifts down from 76% to 74% while third_party roughly doubles, from 6% to 8%.
- Claims backed by a named product or process fall from 34% to 30% while generic, unanchored language rises from 48% to 53%.

- The share of activities left with no identifiable business function climbs from 24% to 33%.

In short: the marginal activity added since the 2023 boom is, on average, less mature and less concretely evidenced than the ones disclosed before it. Volume grew faster than substance.

Disclosed AI activity got bigger in volume but not more concrete, 2021–2025

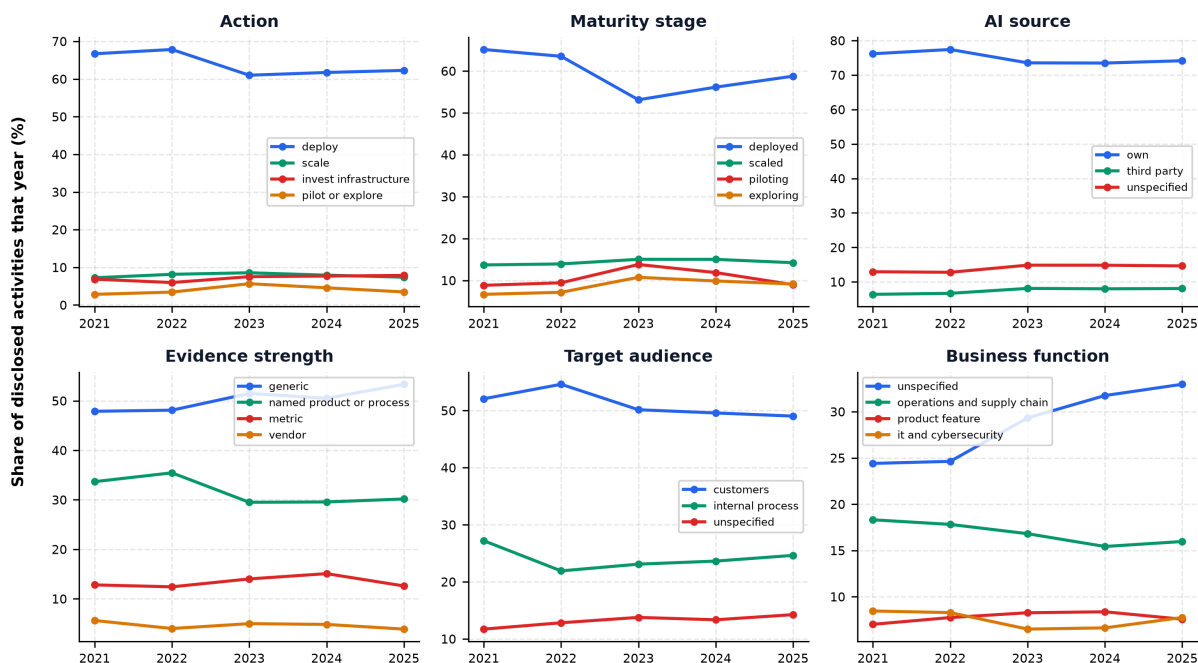


Figure 6: Evolution of disclosed AI activity across five dimensions, 2021–2025 (share of activities per category, by year).

[Table 24](#) (Appendix A) reports the pooled 2021–2026 total for each category (the single number a reader wants for “what share of activities are X, overall”); the figure above is the one that tells the story of how that pool got there.

3.3.2 WHAT FIRMS DISCLOSE DOING: ACTION × OBJECT TYPOLOGY

When free-text objects are standardized into technology families, an empirical typology of corporate deployment emerges: what most S&P 500 firms disclose is deploying an AI product, a predictive model, or a data platform, not building AI from scratch. The top 10 combinations by share of firms are shown below; the full ranking of 19 combinations, with literal textual examples, is in [Table 21](#) (Appendix A).

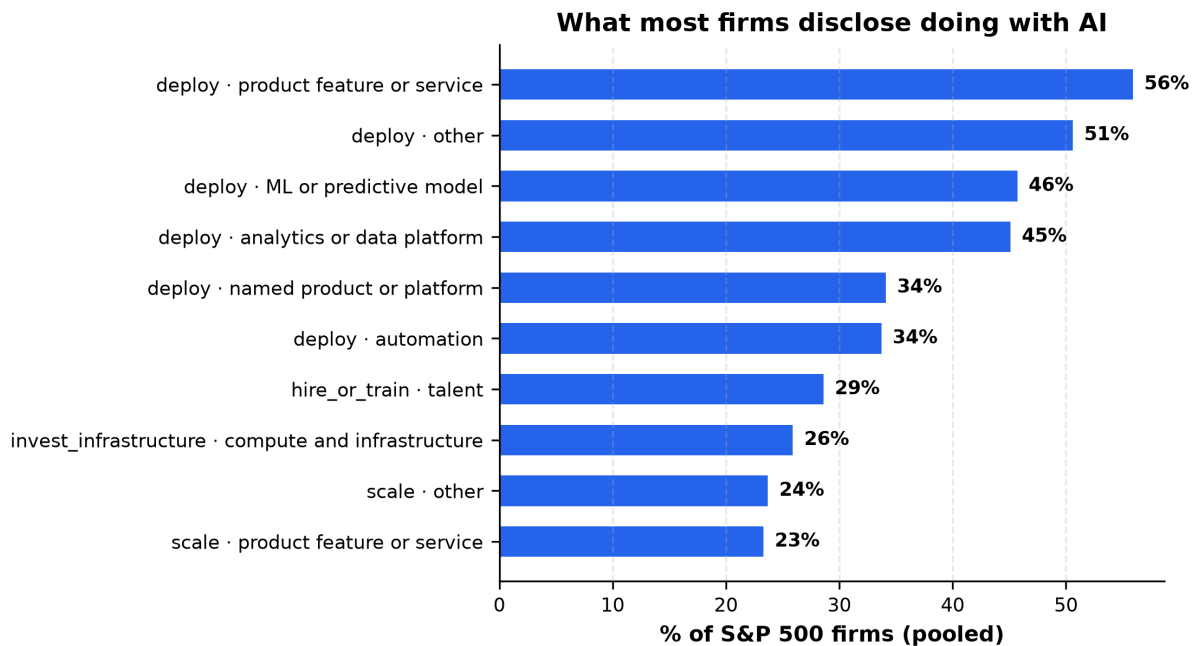


Figure 7: Top 10 action × object combinations by share of S&P 500 firms disclosing at least one such activity.

Intersecting actions, objects, and business functions reveals that the single most prevalent corporate application across the U.S. economy is deploying AI tools to optimize internal operations (30.0% of firms), automating workflows (27.0%), deploying predictive models in supply chains (19.0%), and establishing centralized operational data platforms (18.0%). Customer-facing conversational tools (17.5%) and embedding AI into core sellable products (17.0%) form a secondary cluster, while coding assistants (9.6%), marketing recommendation engines (7.6%), and fraud risk management (8.8%) occupy specialized vertical domains.

3.3.3 TECHNOLOGY PROVENANCE: PROPRIETARY BRANDING VS. VENDOR ATTRIBUTION

A central empirical question in corporate AI is whether public firms are genuine technology creators or commercial consumers of foundation models developed by hyperscalers. Across the corpus, the pipeline extracted 17,193 named entities, classifying each by its operational role.

The distribution of entity roles shows a strong proprietary branding bias:

Table 8: Named-entity roles across the corpus.

Entity role	Instances	Share
Own product or brand	13,333	77.5%
External technology vendor	1,185	6.9%
Strategic commercial partner	780	4.5%
Enterprise customer	728	4.2%
Acquired technology startup	551	3.2%
External foundation model	474	2.8%
Distribution channel	72	0.4%
Competitor or reference	70	0.4%

17,193 named entities total.

Overall, 246 of 510 firms (48.2%) disclose at least one external vendor, commercial model, or technology partner. [Table 9](#) details the external vendor ecosystem cited by S&P 500 corporations.

Table 9: External AI technology ecosystem cited by S&P 500 corporations.

External technology provider / model family	% of Firms citing	Literal named entities extracted
Specialized Enterprise & Data Vendors	41.8%	AMD, Palantir, ServiceNow, Hugging Face, Snowflake
OpenAI / Microsoft Ecosystem	17.3%	OpenAI, Microsoft, ChatGPT, Azure, GitHub Copilot
Google / Alphabet	9.6%	Google, Google Cloud, Gemini, Google AI, Google Vertex AI
NVIDIA	9.0%	NVIDIA, Nvidia, NVIDIA H100, NVIDIA AI software, NVIDIA H100s
Amazon / AWS	6.9%	AWS, Amazon Web Services, Amazon, Amazon AWS, Bedrock
Meta Platforms	3.5%	Meta, Llama, Llama 3, Meta's Llama 3.1, Meta's Llama 2
Anthropic	2.4%	Anthropic, Anthropic, PBC, Anthropic Claude, Anthropic Cloud, Claude
IBM	2.4%	IBM, Watson, Notre Dame — IBM Technology Ethics Lab, watsonx, ibm
Salesforce	1.4%	Salesforce, Salesforce Platform, Einstein

Notes: Vendor citations include entities tagged as `external_provider`, `external_model`, or `partner`. Entities appear in this table only when cited by *other* firms as an external technology source (e.g., Microsoft citing OpenAI, or Lowe's citing Palantir and NVIDIA), excluding self-promotional brand mentions.

Only 5% of all activity statements name an external vendor. Most disclosures leave the technology's origin unstated or attribute it to the firm itself.

While OpenAI and Microsoft represent the dominant external foundation model stack (cited by 17.3% of S&P 500 firms), external vendors are mentioned in only 5.0% of total activity statements. Fully 35.0% of all activity statements explicitly name a proprietary corporate brand. Public companies publicize *what* they achieve with artificial intelligence far more than the third-party infrastructure and foundational models behind those achievements.

3.3.4 SECTORAL DISTRIBUTION OF FRAMES AND ACTIVITIES

The two extraction passes described above (semantic frames and disclosed activities) are shown here purely descriptively, each on its own, crossed against sector (2-digit SIC). No attempt is made here to relate one pass to the other; the aim is only to see where each one concentrates.

Table 10: Frame composition by sector (2-digit SIC, top 10 by firm count): median rates per 1,000 words.

Sector	Firms	Frames/1k	Promo/1k	Risk/1k	Gov/1k
Business services	66	0.196	0.024	0.054	0.02
Instruments	42	0.063	0.003	0.022	0.006
Chemicals	39	0.033	0.002	0.014	0.005
Electric/gas/sanitary	34	0.016	0	0.007	0.004
Electronic equipment	29	0.135	0.015	0.03	0.008
Industrial/computer equip.	28	0.141	0.017	0.029	0.011
Holding/investment offices	27	0.028	0.001	0.012	0.004
Insurance carriers	23	0.038	0.002	0.017	0.01
Depository institutions	21	0.024	0.001	0.01	0.005
Food products	18	0.017	0	0.009	0.003

Table 11: Disclosed activity volume and composition by sector (2-digit SIC, top 10 by firm count).

Sector	Firms	Median activities/firm	Proprietary tech share
Business services	66	101	79.8%
Instruments	42	23	68.0%
Chemicals	39	10	53.5%
Electric/gas/sanitary	34	7	35.1%
Electronic equipment	29	38	84.8%
Industrial/computer equip.	28	54	78.0%
Holding/investment offices	27	9	39.9%
Insurance carriers	23	19	55.6%
Depository institutions	21	16	52.7%
Food products	18	4	38.1%

The two tables are shown side by side deliberately, not merged: [Table 10](#) describes how each sector *talks*, [Table 11](#) describes what each sector *declares doing*, and the two tell related but distinct stories. Business Services and Industrial/Computer Equipment lead on both overall frame volume and activity count: the sectors doing the most AI-related disclosure also report the most concrete activity. But composition differs: risk and governance content together make up 65–69% of the (much smaller) frame volume in Insurance Carriers, Depository Institutions, and Electric/Gas/Sanitary, against 38% in Business Services.

4. DISCLOSURE ARCHETYPES AND THEIR ECONOMIC MEANING

4.1 SIX DIMENSIONS OF AI DISCLOSURE POSTURE

The archetypes are defined from disclosure posture: how a firm frames AI, rather than the activities it reports. The disclosed-activity inventory is reserved for characterizing the resulting archetypes and is never an input to clustering. Because both extractions read the same corporate text, that characterization is descriptive; the external check in this chapter uses XBRL and market data, which do not pass through the LLM pipeline.

Six posture dimensions are computed directly from `gold_ai_frames`, each a simple, named average of a small number of raw indicators, shrunk empirically toward the corpus mean in proportion to how many frames support it (Chapter 2's aggregation rule):

- **Promotional posture:** the average of promotional and strategic-importance rhetoric.
- **Risk orientation:** the share of statements carrying a risk concept.
- **Governance orientation:** the share carrying a governance concept.
- **Temporal posture:** the share of statements describing something already realized, rather than planned, expected, or hypothetical.
- **AI positioning:** the share framed as customer-facing rather than internal.
- **Specificity:** the average of five concreteness flags (named business process, product, vendor, quantified metric, date/timeline).

A seventh dimension, **disclosure intensity** (a percentile rank of AI statements per 1,000 words), distinguishes firms that use the same register at very different levels of disclosure volume.

Table 12: Correlation matrix of the seven posture dimensions.

	Promotional	Risk	Governance	Temporal (realized)	Customer-facing	Specificity	Intensity
Promotional	1	-0.4	0	0.17	0.3	0.27	0.31
Risk	-0.4	1	-0.12	-0.48	-0.48	-0.58	-0.37
Governance	0	-0.12	1	0.24	-0.35	-0.07	-0.2
Temporal (realized)	0.17	-0.48	0.24	1	0.27	0.43	0.21
Customer-facing	0.3	-0.48	-0.35	0.27	1	0.61	0.6
Specificity	0.27	-0.58	-0.07	0.43	0.61	1	0.39
Intensity	0.31	-0.37	-0.2	0.21	0.6	0.39	1

Governance orientation is close to uncorrelated with everything else ($|r| \leq 0.35$, mostly under 0.20): a firm's governance framing says little about its risk framing ($r = -0.12$), its promotional register ($r = 0.00$), or how much it discloses ($r = -0.20$). Risk orientation, by contrast, is the most connected dimension: it correlates negatively with realized/temporal posture (-0.48), specificity (-0.58), customer-facing positioning (-0.48), and intensity (-0.37): a firm that frames AI mostly as a hypothetical risk tends to say less about it, less concretely, and less often about a customer-facing use, all at once.

How this space is diagnosed, not built, by PCA. A principal-components analysis of these seven standardized dimensions is reported as a robustness check, not as the basis for the clustering below: if the first two components captured most of the variance, that would argue for collapsing the space to a two-dimensional summary instead of clustering on all seven.

Table 13: PCA diagnostic on the seven posture dimensions: variance explained, reported as a robustness check, not as the clustering method.

Component	Variance explained	Cumulative
PC1	43.2%	43.2%
PC2	20.2%	63.3%
PC3	12.1%	75.4%
PC4	8.4%	83.8%
PC5	7.0%	90.8%
PC6	5.4%	96.1%
PC7	3.9%	100.0%

The first two components explain 63.3% of the variance (43.2% and 20.2% respectively) and the third still adds a non-trivial 12.1%, not a dominant low-dimensional structure that would justify replacing the seven named dimensions with a two- or three-component summary, but also not so flat that the dimensions are pure noise. This supports the chapter's design choice: keep the six conceptually distinct, managerially interpretable posture dimensions (plus intensity) for clustering, rather than substitute an abstract linear combination whose loadings would need their own translation back into plain language.

4.2 DISCLOSURE ARCHETYPES

Firms with fewer than five AI frames in their pooled SEC filings over 2021–2025 receive the rule-based label **No AI**. This threshold avoids assigning a posture from too little text. K-means clusters all remaining firms on the seven standardized posture dimensions above, and the number of clusters is chosen by resampling each firm's own frames, refitting the entire pipeline (posture rates and

clusters) on the resample, and keeping the largest k whose worst-recovered cluster still clears a 0.60 mean-Jaccard floor against the original assignment.

Table 14: Choosing k by bootstrap stability: mean Jaccard overlap per cluster (25 resamples of each firm's own frames).

k	Jaccard by cluster	Minimum	vs. 0.60 floor
2	0.82, 0.90	0.82	usable
3	0.66, 0.76, 0.81	0.66	usable
4	0.73, 0.63, 0.57, 0.71	0.57	below floor
5	0.44, 0.53, 0.70, 0.61, 0.67	0.44	below floor

$k = 3$ is the largest value clearing the floor ($k = 4$ and $k = 5$ both fail, driven by at least one unstable cluster each).

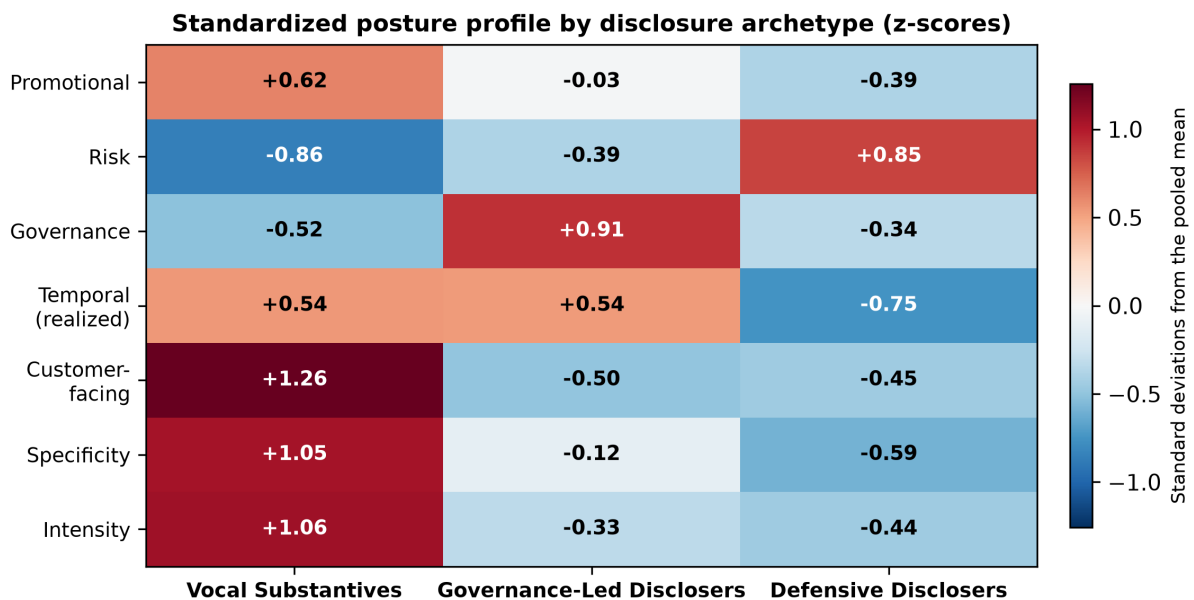


Figure 8: Disclosure archetypes at a glance: standardized (z-score) mean posture profile of the three K-means archetypes.

The three archetypes plus “No AI.” Across the pooled cross-section of 510 firms: **Vocal Substantives** (124 firms, stability 0.76: the highest promotional register, the highest customer-facing positioning, the highest specificity, and the highest disclosure intensity of the three), **Governance-Led Disclosers** (142 firms, stability 0.81: the only archetype defined by above-average governance orientation, with moderate everything else), and **Defensive Disclosers** (191 firms, stability 0.66, the largest archetype: the highest risk orientation and the lowest temporal-realized, customer-facing, specificity, and intensity scores), plus **No AI** (53 pooled firms with fewer than five AI statements in their SEC filings across the sample period). Year-over-year archetype persistence in the panel is 69.3%.

4.2.1 CROSS-PIPELINE CHARACTERIZATION: WHAT THE ARCHETYPES REPORT DOING

None of the three archetypes above was built from any declared-action content: the clustering runs entirely on frame-level communication posture. The disclosed-activity inventory (a separate LLM extraction pass, Chapter 2) is used here to characterize what each archetype reports doing.

This is **not independent validation**: both extractions read the same underlying corporate text, so a firm that discloses more about AI overall has mechanically more opportunity for the semantic-frame pass to find promotional or governance language *and* for the disclosed-activity pass to find a concrete action. A firm's total AI disclosure volume can drive both sides of the comparison at once, without promotional register or governance framing being substantively related to conduct at all. The activity layer is therefore reported here as a descriptive, cross-pipeline characterization, and the comparison below conditions on total AI disclosure volume (log frame count) to reduce this mechanical co-movement before reading anything into the remaining difference.

Table 15: Disclosed behavioral profile by disclosure archetype, from the disclosed-activity inventory (never an input to the posture clustering).

Behavioral feature	Vocal Substantives (n=124)	Governance-Led Disclosers (n=142)	Defensive Disclosers (n=191)	No AI (n=53)
Disclosed activities per firm (median)	96	18	8	1
Customer-facing activity share (%)	53.6%	30.1%	33.1%	33.5%
Technology source: proprietary/own (%)	80.0%	51.7%	50.6%	40.8%
Activities naming an own product/brand (%)	40.7%	17.2%	16.7%	18.3%
Activities reporting a quantified outcome (%)	15.2%	10.4%	10.7%	13.6%
Activities at stage deployed/scaled (%)	74.7%	68.7%	59.5%	58.6%

Notes: Distributions computed from the 27,566-activity inventory across 510 S&P 500 firms, joined post hoc to the posture-based archetype assignment. "No AI" firms with positive activity counts reflect conference-call statements by firms whose formal SEC filings omit AI entirely.

The raw medians in [Table 15](#) are exactly the pattern a shared-source-text confound would produce on its own: the archetype with the most total AI disclosure (Vocal Substantives, Table "Disclosure archetypes at a glance") also reports the most activities. The question worth asking is whether any of that difference survives once total disclosure volume is held fixed.

Table 16: Disclosed activity count by archetype, before and after conditioning on total AI disclosure volume (log frame count). Reference category: Defensive Disclosers.

Archetype vs. Defensive Disclosers	Raw ratio	p	Volume-adjusted ratio	p
Governance-Led Disclosers	1.77x	<0.001	1.44x	<0.001
Vocal Substantives	11.24x	<0.001	2.39x	<0.001
log(1 + AI frames)	-	-	+1.00 (coef.)	<0.001

Total AI disclosure volume explains most of the raw gap (log frame count alone reaches $R^2 = 0.74$ in the adjusted model), exactly as the shared-source-text concern predicts. But a real difference survives conditioning on it: at the same total disclosure volume, Vocal Substantives still report 2.4 times as many activities as Defensive Disclosers ($p < 0.001$) and Governance-Led Disclosers report 1.4 times as many ($p < 0.001$). This is read as a **descriptive cross-pipeline characterization, not independent validation**: the composition statistics in [Table 15](#) (proprietary technology, customer-facing share, deployed/scaled stage) come from the same corpus and carry the same caveat, but the volume-adjusted gap is evidence the three posture archetypes are not simply relabeling "how much a firm discloses."

- **Vocal Substantives** report 80.0% of activities claiming proprietary technology, 40.7% naming a specific own product or brand, 53.6% targeting customers directly, and 74.7% reaching a deployed or scaled stage.
- **Governance-Led Disclosers** show a composition close to Defensive Disclosers (51.7% proprietary, 30.1% customer-facing) despite reporting more activities at the same disclosure volume: governance orientation looks like a distinct communication register layered on top of a similar underlying activity mix, not a proxy for either more or less operational grounding.
- **Defensive Disclosers**, the largest archetype, show the lowest deployed/scaled share (59.5%) of the three and the lowest volume-adjusted activity count, consistent with the risk-heavy, low-specificity register documented in the posture profile above.

4.3 MAPPING THE ARCHETYPES: PROMOTIONAL POSTURE AGAINST DISCLOSED CONDUCT

This section replaces that grid with a single continuous scatter, again carrying the same caveat as the cross-pipeline characterization above: **promotional posture** (from the frame-level posture clustering) on the y-axis, against **disclosed activity volume** (from the disclosed-activity extraction pass) on the x-axis, both read off the same underlying corporate text. Because promotional posture is a *rate* (a share of a firm's own frames), rather than a raw count, it is less mechanically tied to how much a firm discloses overall than a raw activity count is, and the check below confirms the relationship is not simply a volume artifact. Still, the result is reported as a descriptive relationship between two measurements of the same corpus, not as independent confirmation.

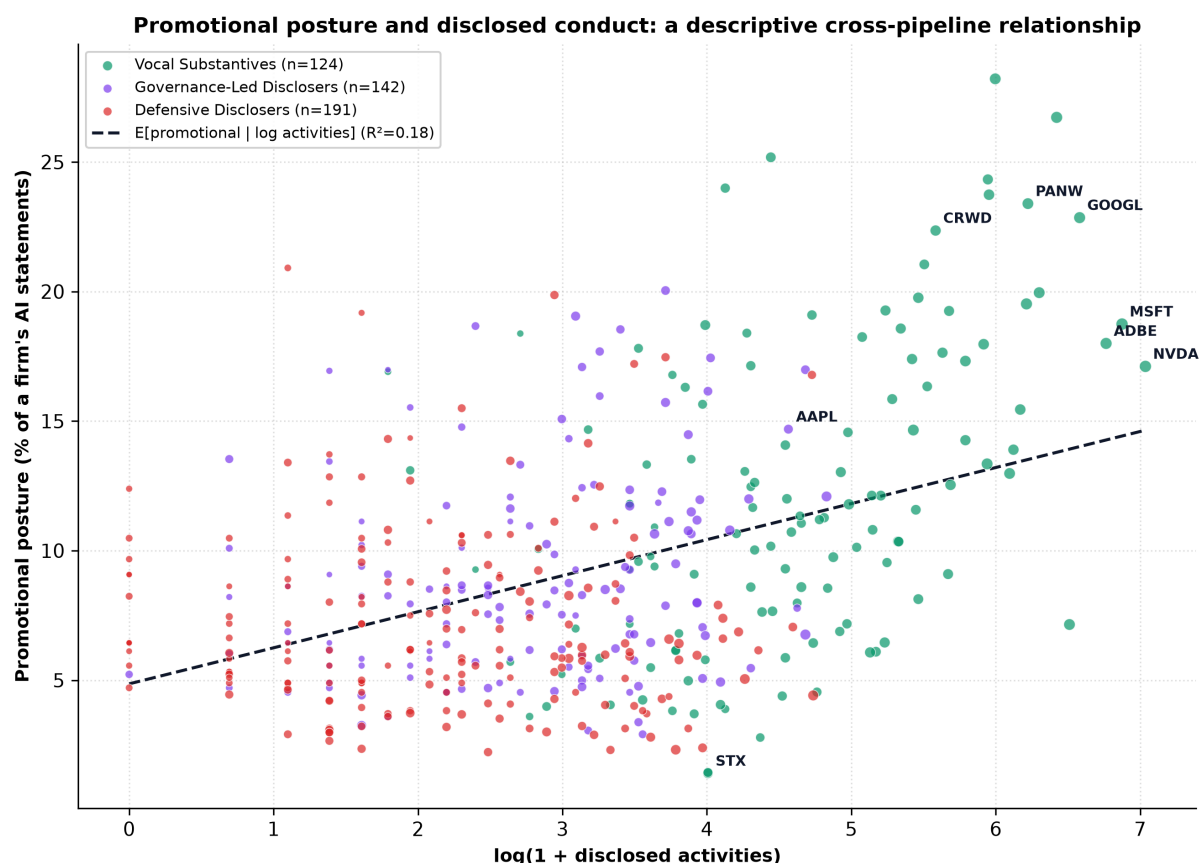


Figure 9: Promotional posture vs. disclosed activity volume (log scale), colored by disclosure archetype. The dashed line is the fitted $\text{promotional_posture} \sim \log(1 + \text{activities})$ relationship. Both axes are read off the same underlying corporate text (semantic frames and disclosed activities are two LLM passes over the same documents), so this is a descriptive relationship, not an independent test.

Promotional posture rises with disclosed activity volume ($\beta = 0.014$ per log-activity unit, $p < 0.001$, $R^2 = 0.176$, $n = 457$). Adding total AI frame count as a control leaves this essentially unchanged ($\beta = 0.012$, still $p < 0.001$, while frame count itself is not significant, $p = 0.32$), so the relationship is not simply “firms with more text score higher on everything,” even though both measurements come from the same corpus.

Firms that disclose more concrete AI conduct also, on average, promote it more, which is consistent with no distinctly “promotional” archetype emerging above: promotional posture is not a separate population of firms, it is a register that scales with disclosed conduct.

The fitted relationship is descriptive only. Its residual is not a measure of AI washing, credibility, or excess promotion; decoupling is measured separately as a disclosure–substance percentile gap.

4.4 ECONOMIC DISTINCTIVENESS

Unlike the activity cross-tabulation above, the financial and market variables used here never passed through the LLM extraction pipeline at all; they come from XBRL filings and market prices, an entirely separate data source from the text the archetypes are built on. This is genuine external evidence, but it answers a narrower question than “are the archetypes real”: it asks whether the archetypes correspond to economically distinct *firms*, not whether the posture dimensions themselves are valid.

An archetype with no economic footprint would still be a real, reproducible communication strategy (Table “Choosing k by bootstrap stability” already establishes that on its own terms); it would just mean that strategy is not one that shows up in fundamentals, which is itself an informative finding, not a failure of the clustering.

The economic-distinctiveness analysis pools all firm-years with filings from 2021 to 2025 (2,475 firm-years, 508 firms; archetype counts: 1,425 No AI, 450 Vocal Substantives, 350 Defensive Disclosers, 250 Governance-Led Disclosers) and tests whether the archetypes correspond to economically distinct firms, both crudely and after sector-by-year adjustment ($X = \alpha[\text{sector} \times \text{year}] + \gamma \cdot \text{Archetype} + \epsilon$, referenced against No AI, firm-clustered standard errors).

Crude medians already put Vocal Substantives ahead of the other two archetypes on the dimensions that matter most for a technology-intensive segment: R&D/sales 12.4% vs. 5.7–6.2% for the other archetypes, gross margin 61.1% vs. 41.5–44.8%, beta 1.09 vs. 0.79–0.81. The residualized panel, within the same sector and year, confirms each archetype separately:

- **Vocal Substantives** is the most consistently distinct archetype: +0.44 log points of market cap ($p < 0.001$), +5.6 percentage points of R&D-to-sales ($p < 0.001$), +11.0 points of gross margin ($p < 0.001$), +0.10 of beta ($p = 0.008$), and +0.97 of price-to-sales ($p = 0.028$), significant on five of the nine dimensions tested.
- **Governance-Led Disclosers** separate from No AI mainly on scale and earnings quality rather than technology intensity: a market-cap premium (+0.33 log points, $p = 0.002$) and a ROIC-minus-WACC premium (+0.065, $p = 0.007$), but *lower* price-to-sales (−0.86, $p = 0.023$) and no R&D or margin premium. This is consistent with the archetype being defined by governance-heavy disclosure that correlates with firm size and regulatory exposure (larger, more heavily regulated firms disclose more governance content) rather than with distinctive AI economics.
- **Defensive Disclosers** barely separate from No AI at all: a small R&D premium (+3.6 points, $p = 0.010$) is the only dimension reaching significance out of nine tested.

No archetype shows a robust advantage in every dimension tested. The story that survives sector-year adjustment is that the archetype most concretely grounded in disclosed conduct (Vocal Substantives, [Table 15](#)) is also the one most consistently distinct on firm fundamentals, not that talking about AI, or emphasizing governance, is by itself.

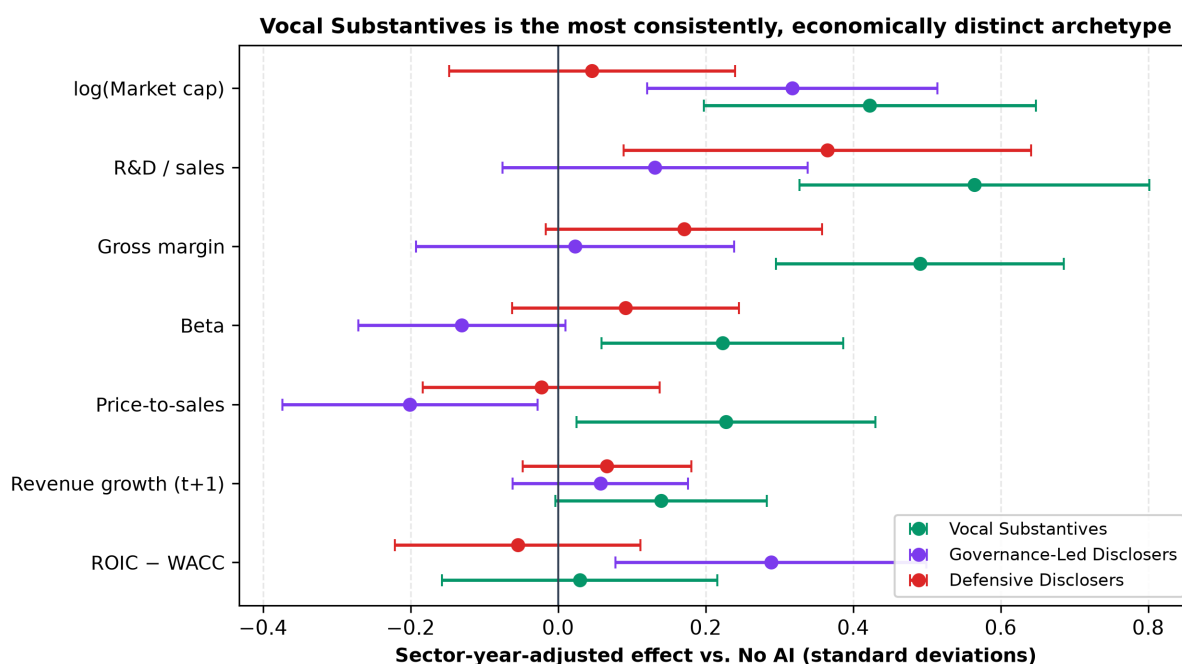


Figure 10: Sector-year-adjusted effect of disclosure archetype on firm fundamentals, relative to No AI (95% CI, firm-clustered SE). Vocal Substantives separates most consistently from the sector-year baseline.

5. DOES AI DISCLOSURE CONTENT CARRY ECONOMIC INFORMATION?

5.1 DOES DISCLOSURE CONTENT ADD ECONOMIC INFORMATION?

The central empirical test of the thesis asks a simple question in a careful way: once you already know how much a firm discloses about AI, and you already know its financial fundamentals, does *what* it says add anything? Five financial outcomes are tested (beta, idiosyncratic volatility, price-to-sales, R&D-to-sales, and next-year revenue growth) starting from the full 2021–2025 panel (2,475 firm-years, 508 firms). Each outcome then drops firm-years missing that outcome or a control variable (R&D/sales, for instance, has much thinner XBRL coverage than market cap) and firm-years whose sector-by-year cell has fewer than two firms, since a fixed effect needs at least two observations to be estimated; the resulting samples, reported in [Table 17](#), range from 649 firm-years (R&D-to-sales) to 1,039 (price-to-sales).

For each outcome, four models are built on top of each other:

- **M0**: a baseline with just financial fundamentals.
- **M1**: the baseline plus how much a firm discloses about AI.
- **M2**: M1 plus seven dimensions of what it discloses (how realized, how deployment-focused, how risk-focused, and so on).
- **M3**: M1 plus which disclosure archetype the firm belongs to, as an alternative to the content dimensions.

The number that matters is how much better M2 predicts the outcome than M1 alone: that gap is the information content adds once volume is already accounted for. That gap is checked three different ways so a single lucky specification can't drive the result: a bootstrap confidence interval, a statistical test on the seven content coefficients, and a permutation test that shuffles the content variables across firms to see what a coincidence would look like.

The content block adds a partial R^2 of 1.8–6.0% across all five outcomes:

- **Beta:** +3.3 points (bootstrap CI [+1.5, +9.7], Wald $F = 3.09$, $p = 0.004$, permutation $p = 0.005$).
- **Idiosyncratic volatility:** +1.8 points (CI [+1.0, +5.7], Wald $p = 0.203$, permutation $p = 0.050$).
- **Price-to-sales:** +2.0 points (CI [+1.4, +8.3], Wald $p = 0.048$, permutation $p = 0.020$).
- **R&D-to-sales:** +6.0 points (CI [+3.4, +15.8], Wald $p = 0.001$, permutation $p = 0.005$).
- **Next-year revenue growth:** +5.2 points (CI [+0.9, +13.2], Wald $p < 0.001$, permutation $p = 0.005$).

Every bootstrap interval excludes zero, adjusted R^2 rises with the content block in all five outcomes, the Wald test rejects at 5% in three of five (idiosyncratic volatility falls short at $p = 0.203$), and the sector-by-year permutation, which destroys only the content-outcome association while preserving sector, year, fundamentals, and volume, rejects in all five. The magnitude is modest but real: of the same order as what disclosure volume itself contributes (0.2–5.6 points) and larger than what segment membership contributes as a dummy (0.1–1.7 points).

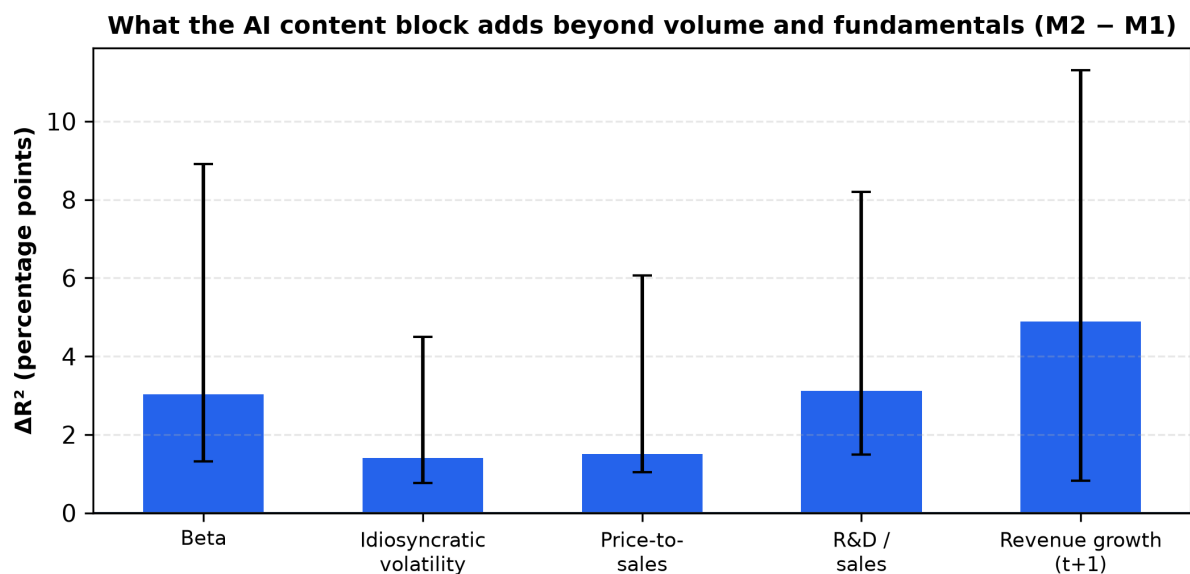


Figure 11: Incremental explanatory power of AI disclosure content beyond volume and fundamentals: partial R^2 by outcome, with bootstrap 95% CI (firm-clustered).

The coefficient plot below shows the seven M2 content dimensions separately. Specificity is positive and statistically distinguishable from zero for idiosyncratic volatility and next-year revenue growth, but it is not individually precise in every outcome. Promotion is not a repeated positive signal; its only individually precise coefficient is negative for next-year revenue growth. The joint block results above, rather than any single coefficient, are therefore the primary evidence on incremental information.

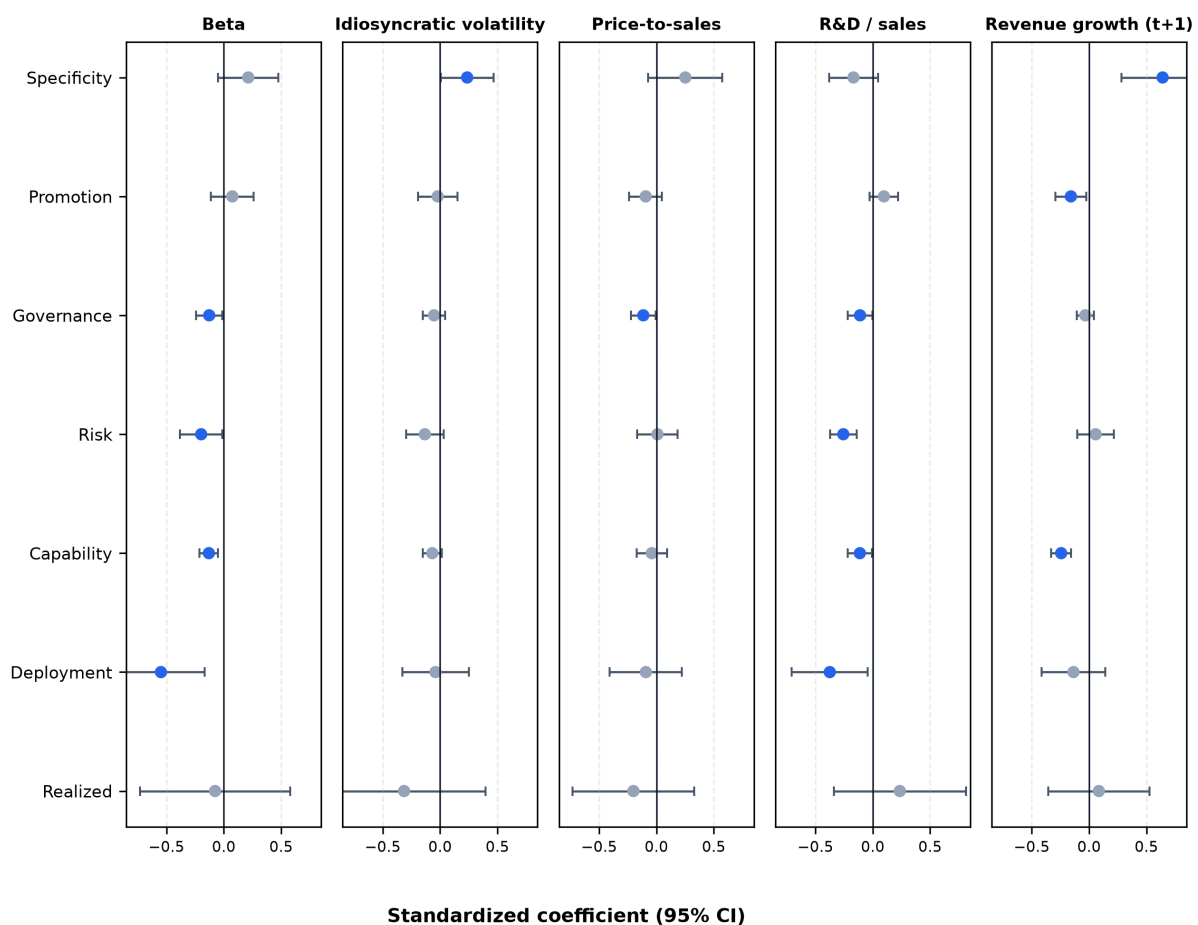


Figure 12: M2 standardized coefficients by outcome, with 95% confidence intervals from firm-clustered covariance estimates. Each model includes financial controls, AI disclosure volume, and sector-by-year fixed effects.

Taken jointly, the pattern is consistent with, but does not establish, the signalling and disclosure-tone mechanisms discussed by Spence and Rogers et al. (Spence 1973, 363–64; Rogers, Van Buskirk, and Zechman 2011, 2155–56).

5.2 ROBUSTNESS CHECK: MIX OR VOLUME?

Re-expressing the content block as within-firm semantic shares (each concept's frames as a fraction of the firm's total AI frames, with volume entered separately) rather than log-intensities halves the ΔR^2 in idiosyncratic volatility, price-to-sales, and growth and the Wald test stops rejecting in those three outcomes, while beta and R&D-to-sales are essentially unaffected. Part of the content signal is therefore the intensity of each dimension and not only the mix; the compositional signal alone survives only where systematic risk and R&D are concerned.

5.3 WHAT VS. HOW

Six activity-count families (customer-facing deployment, internal deployment, own AI, named provider, infrastructure, quantified outcome, log-transformed intensities with zeros) are entered as an alternative content block and as an additional block layered on top of the semantic dimensions. The two blocks are not redundant: together they raise ΔR^2 to 2.1–7.8 points across the five

outcomes, and each block retains at least half its individual contribution once the other is already included, with two informative exceptions:

- **Idiosyncratic volatility:** activity absorbs most of the semantic signal (semantics given activity $\Delta R^2 = +0.009$, $p = 0.50$). Idiosyncratic risk tracks what a firm does more than how it narrates it.
- **Revenue growth:** activity contributes roughly 2.4 times what semantic style contributes once each is conditioned on the other ($+0.029$ given semantics vs. $+0.012$ given activity). Growth is anticipated more by the concrete activity a firm describes than by its rhetorical specificity, though both remain jointly informative.
- **Beta and price-to-sales:** both blocks hold up independently, each still significant conditional on the other.

The joint reading: specificity of language matters in addition to disclosed activity, not as a substitute for it, and where the outcome is fundamentally about “doing,” such as idiosyncratic risk and growth, the activity block dominates.

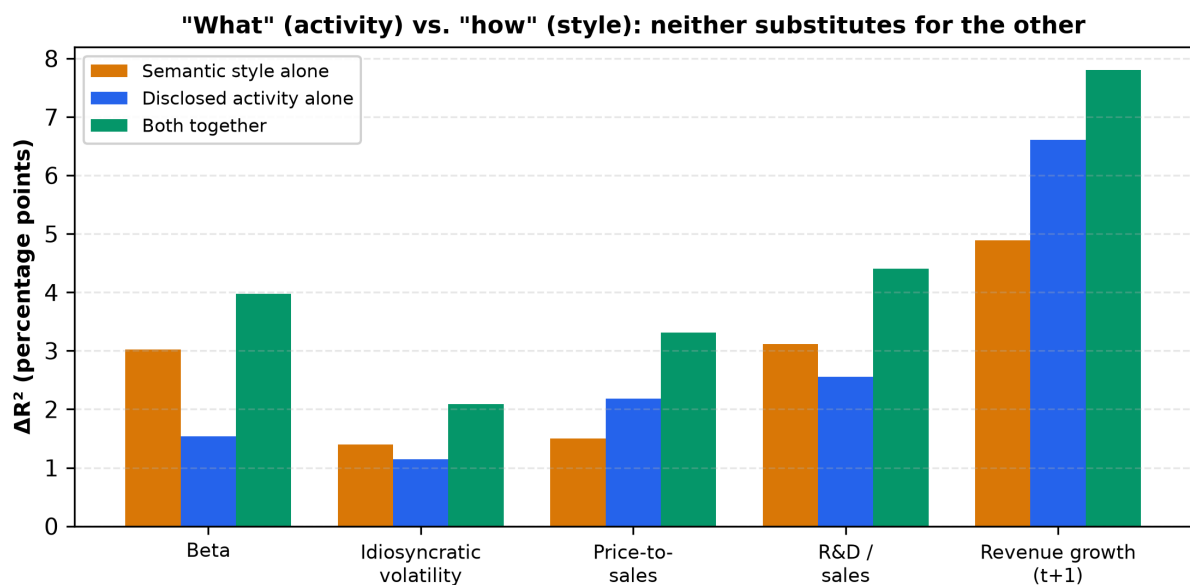


Figure 13: What vs. how: ΔR^2 contributed by semantic style alone, disclosed activity alone, and both together, by outcome.

[Table 17](#) reports the central empirical test of whether textual AI disclosure carries incremental economic information beyond disclosure volume and financial fundamentals. Across five financial outcomes, Model 1 (M1) includes firm size, sector-by-year fixed effects, disclosure volume, and standard financial controls. Model 2 (M2) adds the seven semantic frame dimensions (not disclosed activity, added separately as its own block below). In every case, semantic content adds statistically significant explanatory power (partial R^2 of 1.8 to 6.0 percentage points, bootstrap intervals excluding zero), rising to 2.1–7.8 points once disclosed activity is layered on top of the semantic block.

Table 17: Central incremental-information results.

Outcome	n	R2 M1	R2 M2	DeltaR2 content	95% bootstrap CI	Partial R2	Wald p	Permutation p
Beta	1,037	0.095	0.125	0.03	[+0.013, +0.089]	3.3%	0.004	0.005
Idiosyncratic volatility	1,037	0.211	0.225	0.014	[+0.008, +0.045]	1.8%	0.203	0.05

Outcome	n	R2 M1	R2 M2	DeltaR2 content	95% bootstrap CI	Partial R2	Wald p	Permutation p
Price-to-sales	1,039	0.235	0.25	0.015	[+0.010, +0.061]	2.0%	0.048	0.02
R&D/sales	649	0.479	0.51	0.031	[+0.015, +0.082]	6.0%	0.001	0.005
Revenue growth (t+1)	1,003	0.067	0.116	0.049	[+0.008, +0.113]	5.2%	<0.001	0.005

6. CREDIBILITY AND DECOUPLING

This chapter brings together the two forms of divergence relevant to credibility. The first is **within-disclosure decoupling**: a firm-year's gap between AI disclosure intensity and the concrete, grounded activity it discloses. The second is **cross-venue decoupling**: the difference between what the same firm says in statutory filings and in earnings calls. These measures answer related questions but are not interchangeable; the SEC exercise is assessed as a secondary test of whether either pattern visibly changed after 2024 scrutiny.

6.1 WITHIN-DISCLOSURE DECOUPLING

The AI-washing index is the percentile-rank gap between disclosure intensity and disclosed substance:

$$W_{it} = \text{pctrank}(\text{Disclosure}_{it}) - \text{pctrank}(\text{Substance}_{it}), \quad \text{Substance}_{it} \\ = \log(1 + n_{\text{activities},it}) \times \text{Grounding}_{it}.$$

It is defined for the 1,449 firm-years that disclose AI at all. The distribution has face validity: its positive tail contains disclosure-heavy firm-years with zero or near-zero grounded activity, rather than firms with the greatest absolute activity volume. It is also persistent within firm (year-over-year Spearman $\rho = +0.602$; $n = 929$ pairs), while remaining distinct from a fixed firm trait.

Its economic role is more limited. The next section tests whether it adds information beyond disclosure volume, semantic content, and disclosed activity. Its full construction, distribution, and validation are reported in Appendix C.

6.2 DOES WITHIN-DISCLOSURE DECOUPLING ADD ECONOMIC INFORMATION?

The test adds W after disclosure volume, semantic content, disclosed activity, financial controls, and fixed effects. This asks whether the residual gap between disclosure and disclosed substance adds information once the underlying components are already observed.

It does not. For R&D-to-sales, the outcome closest to technological intensity, adding W raises R^2 from 0.6032 to 0.6033 ($n = 445$; $p = 0.896$). The same result holds for beta, idiosyncratic volatility, price-to-sales, and next-year revenue growth ($\Delta R^2 \leq +0.0007$; $p \geq 0.55$ in every case). The credibility gap is therefore stable and interpretable, but it is not an additional economic signal beyond specificity and concrete disclosed activity.

6.3 CALLS VS. FILINGS

The most tightly identified result in the thesis compares the same firm, in the same fiscal year, across the two venues that differ most in liability exposure: the earnings call, an informal and voluntary disclosure venue (Frankel, Johnson, and Skinner 1999, 133–34; Matsumoto, Pronk, and Roelofsens 2011, 1383–84), and the SEC filing, a monitored venue carrying legal liability for material misstatement.

The design uses firm $\times$ fiscal-year fixed effects with exactly two channel observations per cell, so the estimator reduces to a within-cell difference: call intensity minus filing intensity. This cancels

everything common to the firm in that period, including the industry-wide AI boom itself. Because venue bundles legal exposure with differences in audience, format, length, and topic selection, the estimate is interpreted as a within-firm, within-year venue-associated difference, not as the isolated causal effect of legal liability.

The extensive-margin sample comprises 2,281 firm $\times$ fiscal-year cells across 479 firms (449 present on both sides of the 2024 cutoff), aligned by the fiscal period each document actually covers rather than its filing date. Of these cells, 1,021 contain calls that mention no AI at all (5,238 of 7,947 transcripts).

The denominator matters here: a scorable call paragraph runs to 129 words on average against 67 for a filing paragraph, so a raw per-paragraph count would overstate the gap by roughly half. This thesis reports the rate per 1,000 words instead. On that basis, the same firm, in the same fiscal year, places 9.9 $\times$ as many AI-promotional statements in its earnings call as in its filing, and 17.1 $\times$ as many quantified claims, even though the filing has a higher share of documents mentioning AI at all (65% vs. 55%). Chapter 3 already showed that what firms added to their filings between 2023 and 2025 was, above all, an AI risk factor and a governance line: this cross-venue result is consistent with that pattern, and suggests the product pitch is being made somewhere else.

[Table 18](#) and [Figure 14](#) report the same comparison two ways: the table gives the channel means and the call-to-filing ratio, and the figure gives the within-cell difference with a firm-clustered 95% bootstrap confidence interval.

Table 18: Channel means and call-to-filing ratio, per 1,000 words.

Measure (per 1,000 words)	Filing	Call	Call / Filing
Promotional claims	0.006	0.054	9.9 $\times$
Quantified claims	0.003	0.052	17.1 $\times$

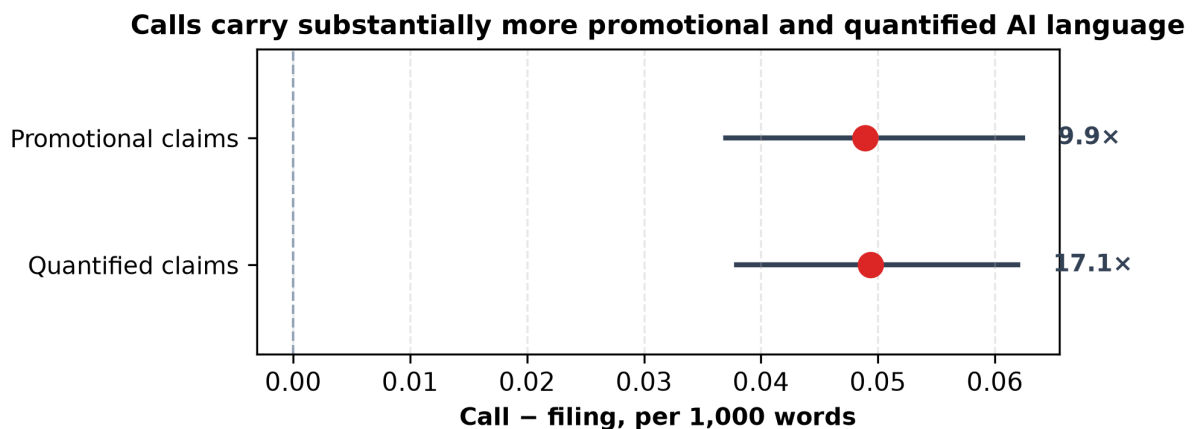


Figure 14: Within-firm, within-year call-minus-filing difference in AI disclosure rate, with a firm-clustered 95% bootstrap confidence interval. Ratios (call / filing) are annotated on the right.

Section 6.2 provides an independent, denominator-free check by comparing the same firm's *disclosed activities* rather than word or paragraph counts across venues, and finds the activity mix itself essentially unchanged while what is attached to it (names, figures, providers) differs sharply, the most robust reading of this gap. This is a promotional gap between venues, not evidence that any individual claim is false. The SEC's Welltower letter provides a motivating instance in which staff referenced the filing, call, and press release together (U.S. Securities and Exchange Commission 2025,

1); evidence on SEC comment letters more generally shows that the process can prompt changes in future disclosure (Bozanic, Dietrich, and Johnson 2017, 1–5).

6.4 ACTIVITIES ACROSS VENUES

Comparing disclosed-activity composition across 459 firm × fiscal-year cells (183 firms with at least three activities in each channel), the *mix* of what firms say they do is essentially identical across venues: customer-facing deployment, internal deployment, and own-AI shares differ by only 0.6 to 2.2 percentage points, none distinguishable from zero at 5%.

What differs is the evidence attached to the same claim. The call adds:

- a declared business function (+11.7 points);
- a quantified outcome (+11.0 points; the filing almost never states the figure);
- a named external provider or partner (+5.3 points, nearly triple); and
- a named product (+3.8 points),

while the filing retains relatively more infrastructure investment, talent, and governance language. The gap is not “different activities depending on the audience” but “the same activity, with figures and names for the analyst and without them in what is signed,” a pattern stable across fiscal years, with the quantified-outcome gap ranging from +9.0 to +11.8 points between 2021 and 2025.

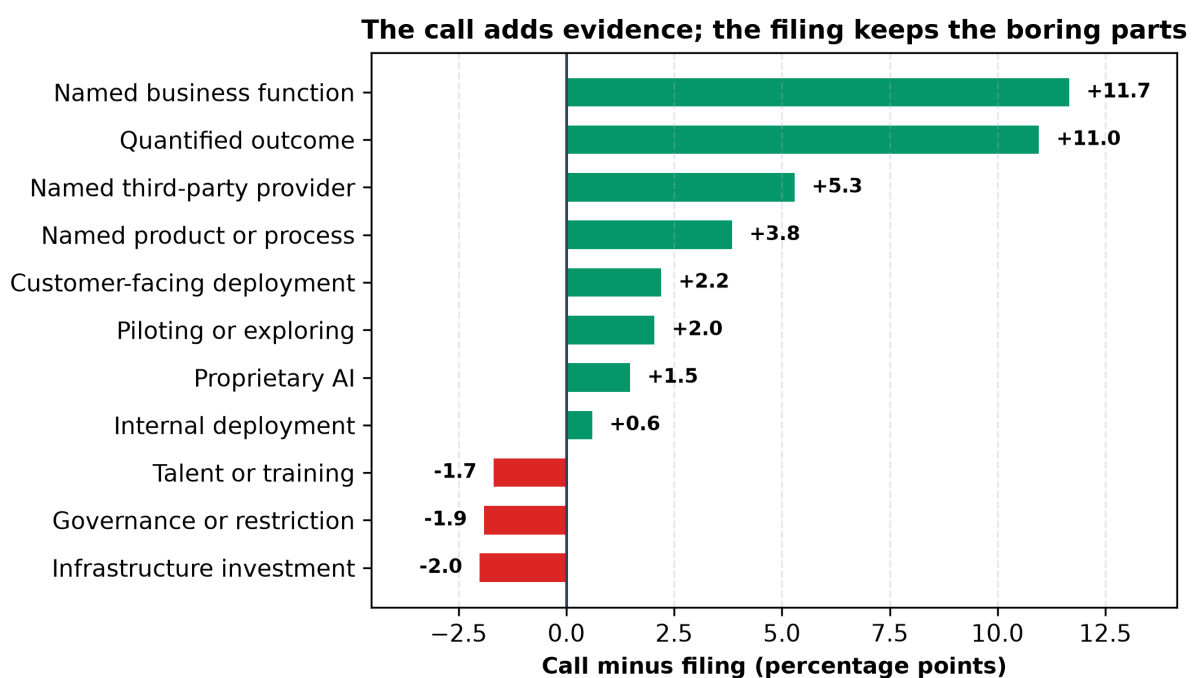


Figure 15: Same activity, different evidence: call-minus-filing gap in how often each activity carries a given piece of evidence, ordered from largest to smallest (percentage points).

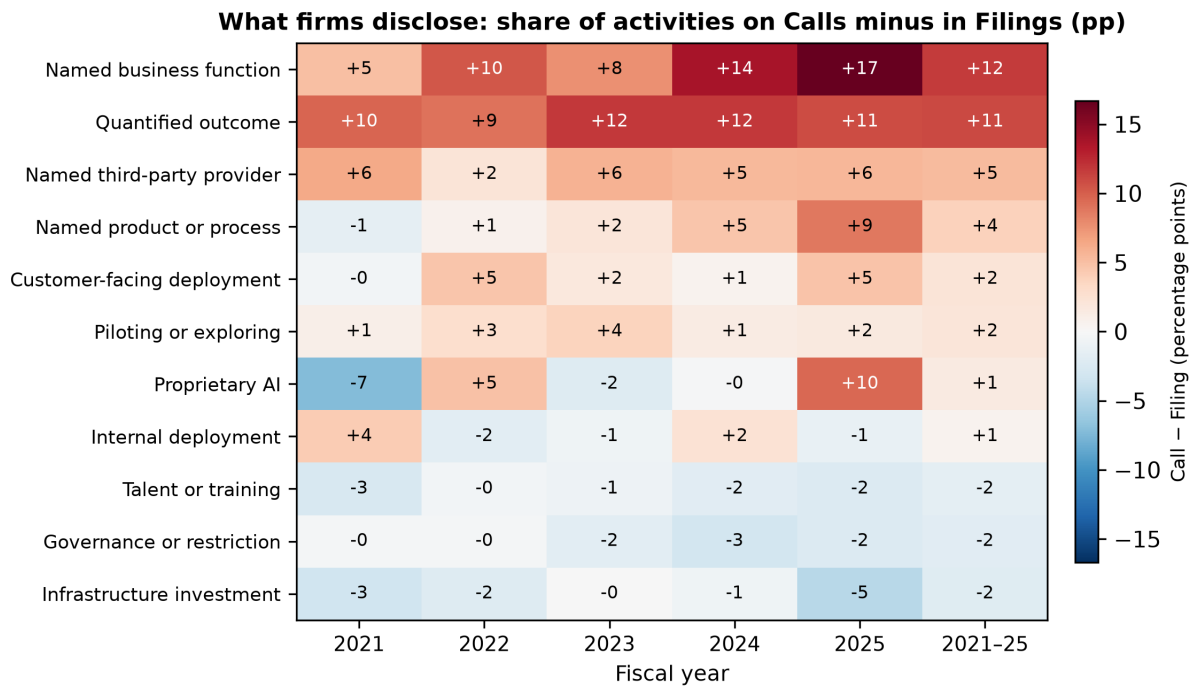


Figure 16: Cross-venue activity gap: comparison of disclosed AI activity frequencies, concreteness, and evidence grounding between formal SEC filings (10-K/10-Q) and earnings conference calls.

6.5 VENUE GAP OVER TIME

With firm fixed effects on the gap itself, the promotional gap (per 1,000 words) widens from a flat base of 0.019 in 2021–2022 to 0.054 in fiscal 2023, 0.077 in 2024, and 0.080 in 2025 (post-2024 coefficient +0.047, $p < 0.001$, on the same 449 firms present on both sides of the cutoff). Governance is the one dimension where the gap does not grow: filings add governance content at roughly the same rate calls add promotion, so the difference nets to zero.

Pre-trend tests fail everywhere, and they have to: the gap was already widening in fiscal 2023, a full year before the SEC’s public warning. So the SEC’s 2024 scrutiny is not separably identifiable against this gap. A rate-normalized check (conditioning on cells where both channels discuss AI, so the comparison is not confounded by the extensive margin) confirms the same null: the post-2024 coefficient on the conditional promotional rate is +0.015 ($p = 0.36$), with flat pre-trends.

Firm-level trajectories vary in level but not in this general pattern:

- NVDA’s gap widens from +0.29 to +0.44 since 2023.
- PLTR stays persistently above +0.3.
- HPE is a rare case where the gap closes, from +0.25 to –0.03.
- Welltower itself, the case motivating the design, has too little AI-call content before 2025 to estimate a firm-specific trajectory.

The firm-level gap does not correlate with the within-disclosure AI-washing index (Spearman +0.034, $n = 171$ firms scoreable on both measures) — a reminder that these are different forms of divergence, one measured within a filing and one measured across venues, not two readings of the same underlying quantity.

6.6 SEC 2024

To test this, every firm-quarter (11,302 of them, across all 510 firms) is scored on how much AI exposure it had before 2023, and the design asks whether firms with more pre-existing exposure changed differently around the SEC's warning than firms with less.

The standard check for this kind of design is whether the two groups were already on different trends *before* the event: if they were, any “effect” you find afterward could just be that pre-existing trend continuing, not something the SEC caused. Here, that check fails for every outcome tested (promotional language, risk, hypothetical framing, total AI volume, specificity) because more-exposed firms were already pulling away from less-exposed firms well before December 2023. The design cannot separate a scrutiny effect from a trend that was already there.

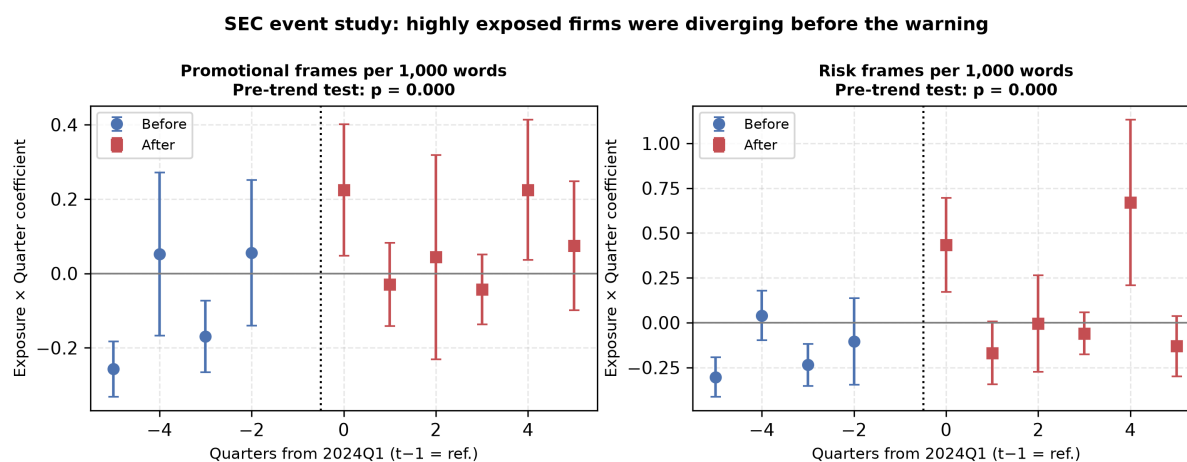


Figure 17: Diagnostic event study: SEC regulatory sweep and enforcement inquiries around AI disclosures across event quarters, showing pre-trend divergence.

A simpler difference-in-differences design (108 more-promotional firms vs. 382 controls, split by pre-period promotional intensity) finds that the more-promotional group added more governance and specificity after December 2023 than the control group (+0.0084 and +0.0161 per 1,000 words, $p = 0.001$ and $p < 0.001$), consistent in direction with a scrutiny response; pre-trends for both outcomes are clean ($p = 0.138$ and $p = 0.245$), though the same-dimension reference outcome (promotional intensity itself) has a pre-trend that sits at the margin of significance ($p = 0.053$), and the jump concentrates in the 2024 filing season rather than tracking the enforcement timeline precisely.

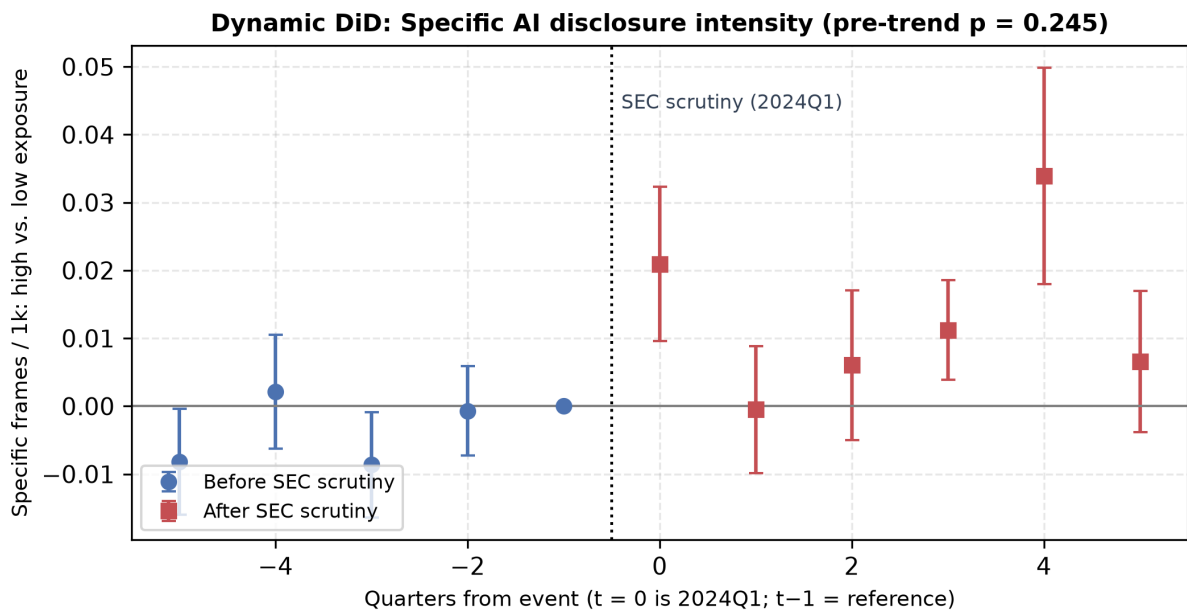


Figure 18: Difference-in-differences dynamic trajectory: specific disclosure intensity per 1,000 words following SEC regulatory scrutiny.

Identification verdict: the causal effect of SEC scrutiny cannot be separated from the concurrent generative-AI boom with this design. What is observed after 2024 (more risk disclosure, more governance, a shrinking promotional share in Chapter 3) is equally consistent with a scrutiny response and with what any firm adds to its 10-K once a technology becomes material.

One sentence on the second shock examined: DeepSeek was also tested (a January 2025 cutoff, the same event-study design, 5,287 firm-quarters), but no discontinuity separable from prior trends was identified; the full analysis, including the failed pre-trend tests, is reported in Appendix E.

Firm-level trajectories and event-study specifications across alternative outcomes are detailed in Appendix E.

Table 19: SEC identification verdict.

Design	Pre-trend test	Verdict
Event study, continuous exposure, firm + quarter FE	Fails for all 5 outcomes ($p < 0.001$)	Not identifiable
Simple DiD, promotional split	At the margin ($p = 0.06\text{--}0.09$); jump concentrated in 2024 10-K season	Not identifiable as a scrutiny effect
Within-firm channel gap (call – filing), post-2024	Fails ($p < 0.01$); gap was already widening in FY2023	Gap persists; no closure after scrutiny
DeepSeek event study (Appendix E)	Fails for 4 of 5 outcomes	Not identifiable

7. DISCUSSION AND MANAGERIAL IMPLICATIONS

The results support three complementary interpretations. Specific, falsifiable claims fit a signalling account; the disappearance of the no-AI segment and the limited economic distinctiveness of most archetypes fit institutional diffusion; and the widening call–filing gap is consistent with cheap talk in a more discretionary, lower-formality disclosure venue. These are associations, not evidence that one mechanism causally governs all AI disclosure.

7.1 ISSUER IMPLICATIONS

The measurement built across this thesis converts directly into a disclosure checklist. Credible AI disclosure:

- names a concrete activity, not “AI capabilities,” which 86% of firms use as an empty placeholder object;
- states a clear stage (deployed or scaled, not merely explored or piloted);
- names a specific product or process;
- reports a quantified outcome;
- discloses the source of the AI, including provider and partner relationships, since 16% of activities in the corpus leave this unstated, and undeclared origin is far more common among Defensive Disclosers and Governance-Led Disclosers (33% each) than among Vocal Substantives (10%); and
- is consistent between what is said on the earnings call and what is filed with the SEC. The Welltower comment letter illustrates why that comparison can be useful: SEC staff referenced statements across the filing, call, and press release, without finding a violation or defining a general consistency test.

7.2 INVESTOR IMPLICATIONS

Of the textual dimensions available in AI disclosure, specificity carries incremental economic information (Chapter 5: 1.8–6.0 points of partial R^2 in beta, idiosyncratic volatility, price-to-sales, R&D-to-sales, and next-year revenue growth), while promotional tone carries essentially none. Disclosed activity adds information beyond semantic style, especially for operating risk and growth.

Investors should read earnings-call AI claims against what the same firm has filed: the same activity mix is described in both venues, and the call does supply concrete evidence the filing typically omits (quantified outcomes, named providers, named products) alongside more promotional language, so cross-venue consistency, not the call’s promotional tone by itself, is the more useful credibility signal. This is a gap that has been widening since the 2023 AI boom and has shown no sign of narrowing since the SEC’s December 2023 warning.

The within-disclosure index is best read through decoupling theory: formal disclosure can run ahead of the concrete activity a firm discloses, without establishing what the firm actually does. The cross-venue gap is the closer analogue to the mechanism the SEC has enforced. Neither measure has an external operational benchmark, and the SEC design does not identify a causal effect of scrutiny.

Table 20: Issuer / investor checklist.

Dimension	Issuer guidance	Investor read
Named activity	State the specific action-object, not “AI capabilities”	Absence is the single most common gap (86% of firms)
Stage	Report deployed/scaled, not only piloted	Piloting-only claims should be discounted relative to deployed claims
Product/process named	Name it	Named claims carry more of the incremental economic signal
Quantified outcome	Report the figure in the filing, not only on the call	The filing omits this far more often than the call (61%→37% pooled-to-filing-only)
AI source / provider	Disclose whether AI is own, third-party, or co-developed	16% leave this unstated; undeclared origin clusters in the least-grounded segment

Dimension	Issuer guidance	Investor read
Venue consistency	Match filing content to call content	The 9.9× promotional gap has not closed since 2024 scrutiny

8. CONCLUSIONS

This thesis follows a single progression: AI talk expanded rapidly; its content differs systematically across firms; concrete specificity and disclosed activity contain the useful economic information; and credibility gaps emerge most clearly across venues.

First, AI disclosure grew nearly seven-fold in intensity and reached 92% of S&P 500 firms between 2021 and 2025. Firms do not merely differ in volume: their statements organize into reproducible disclosure profiles, but only Vocal Substantives—those reporting the greatest concrete activity even after conditioning on volume—show a broad, sector-year-adjusted economic distinction.

Second, semantic content and disclosed activity add modest but robust information beyond disclosure volume and financial fundamentals. The signal is concentrated in specificity and concrete activity, not promotional tone. The within-disclosure washing index is stable and interpretable as a gap, but adds no incremental economic information once those underlying components are in the model.

Third, credibility is institutionally contingent. The same firm-year reports a similar activity mix in filings and calls, yet calls attach substantially more evidence and promotion to those activities. This cross-venue gap widened with the 2023 boom and shows no detectable closure after the SEC’s 2024 scrutiny; the available design does not identify a causal effect of enforcement.

The methodological contribution is a five-layer, largely reproducible extraction pipeline that recovers structured, falsifiable claims and grounded activities from over nine million paragraphs at a cost no purely human-coded design could match, with venue-specific measurement error documented rather than assumed away. The managerial contribution is a concrete disclosure checklist for issuers and a cross-venue credibility framework for investors, both grounded in what the measurement shows carries information.

Theoretically, the results are most consistent with a signalling account of specificity, a legitimacy/institutional-diffusion account of the “no AI” segment’s disappearance, and a cheap-talk account of earnings calls as a more discretionary, lower-formality disclosure venue. The main claims of this thesis are measurement and association rather than causal effects: the most tightly identified design isolates a venue-associated disclosure difference, not the causal effect of legal liability, and the SEC and DeepSeek shock designs fail pre-trend tests and are reported as failed identification rather than assigned a causal reading.

Four limitations are carried through every chapter:

1. Every frame and activity label comes from an LLM judge: the measured reliability (κ 0.81–0.87) is agreement between two LLM judges, not accuracy against an independent ground truth.
2. “Disclosed behaviour” and “disclosed activity” measure what a firm says it does, not what an audited operational record would confirm.
3. The SEC and DeepSeek shocks are not identifiable against the background AI boom with the designs available, a null result reported rather than concealed.

4. The AI-washing index has no external benchmark: it compares a firm's own disclosure to its own disclosed activity, both extracted from the same corpus, and cannot check either against an audited operational record.

Cross-venue results are robust to using words rather than paragraphs as the denominator and to denominator-free activity comparisons.

Future research should extend the cross-venue design to a synthetic-control or matched-donor-pool setting if a denser call-transcript series becomes available, particularly for cases like Welltower with too little pre-period call content to estimate a firm-specific trajectory; and extend the comparative international panels (Chile, Italy) noted in the data section to test whether the venue-gap and segmentation results generalize outside a single, SEC-regulated regime.

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APPENDICES

APPENDIX A: ANALYTICAL STACK

The pipeline and analysis were implemented in Python. Parquet files are the primary storage format and DuckDB provides queryable views over those files. pandas and NumPy handle panel construction and numerical transformations; scikit-learn, SciPy, and statsmodels support the prefilter, clustering, diagnostics, and cluster-robust regressions; and matplotlib produces the figures.

Text embeddings use the Hugging Face model BAAI/bge-m3 (1,024-dimensional dense vectors) for semantic-anchor features and retrieval. The principal LLM judge is qwen/qwen3.7-flash, called through OpenRouter with structured outputs via pydantic-ai, for semantic-frame and disclosed-activity extraction. Gemini is used only as an independent judge in the agreement checks, not as the production extractor.

GPU embedding and document-processing workloads were run on a Vast.ai instance with an NVIDIA RTX 5090, using a custom-built mrbungie/datascience-runtime image. Other analytical tasks ran locally on a MacBook Pro with an M2 Pro chip. Code was synchronized with Git and data artifacts with Backblaze. SEC filings and XBRL facts are collected through EDGAR tooling. Part of the financial and market data, including historical prices, was obtained through yfinance; Fama–French factors supply the market and risk-free-rate series used in the risk measures.

APPENDIX B: DATA PIPELINE

The data pipeline has five stages: collect and segment filings and calls; deduplicate scorable text; identify AI-relevant text with the prefilter; extract semantic frames; and extract firm-specific disclosed activities. Each stage writes a parquet artifact consumed by the next, while the DuckDB file exposes views over the current parquet data rather than a second copy. The current corpus totals in Chapter 2 and the form-by-form table are generated from the processed paragraphs table at render time.

APPENDIX C: SUPPLEMENTARY DATA

The full ranking of action × object disclosed corporate activities in the S&P 500 ([Table 21](#) below) complements the top-10 chart already shown in Chapter 3 (“What firms disclose doing”).

Table 21: Full action x object ranking, pooled vs. filings-only share of firms, with literal textual examples.

Action x Object family	% Firms (Pooled)	% Firms (Filings only)	Literal textual examples
deploy · AI, unspecified object	88.1%	81.7%	
deploy · product feature or service	60.5%	41.6%	ai engine, interconnect products, advertising tools, ai software
deploy · other	54.8%	32.3%	on-device ai, edge ai, aip, ai networks
deploy · ML or predictive model	49.5%	27.0%	ai models, machine learning models, ai algorithms, machine learning model
deploy · analytics or data platform	48.8%	30.4%	ai platform, ai platforms, expert platform, ai-powered analytics
pilot_or_explore · AI, unspecified object	42.0%	26.1%	
scale · AI, unspecified object	38.6%	26.8%	
deploy · named product or platform	36.9%	23.1%	aiops, aip, embedded ai, mist ai
deploy · automation	36.5%	19.1%	automation, ai and automation, automation tools, automation and ai
invest_infrastructure · AI, unspecified object	34.4%	25.3%	
hire_or_train · talent	31.0%	26.8%	ai talent, talent, employees, workforce
integrate · AI, unspecified object	28.9%	21.7%	
invest_infrastructure · compute and infrastructure	28.0%	17.4%	ai infrastructure, data centers, compute capacity, cloud infrastructure
scale · other	25.7%	10.0%	ai integration, hybrid cloud and ai strategy, edge ai, ai portfolio
scale · product feature or service	25.3%	9.6%	digital capabilities, data ecosystem, efx.ai capabilities, firefly services
measure_outcome · other	23.1%	10.6%	ai orders, ai demand, business performance, inference performance
develop · AI, unspecified object	22.5%	18.9%	
measure_outcome · financial outcome	21.2%	10.4%	ai revenue, productivity, revenue, productivity gains
deploy · compute and infrastructure	21.0%	13.2%	ai infrastructure, ai pc, ai accelerators, ai accelerator
invest_infrastructure · other	20.8%	11.5%	ai research and development, research and development, hybrid cloud and ai strategy, ai research

Notes: Pooled sample covers 510 S&P 500 firms with filings and call transcripts. “Filings only” restricts the denominator to statutory SEC filings (10-K, 10-Q, DEF 14A, 8-K), illustrating the significant contribution of earnings calls to corporate disclosure of commercial tools, partnerships, and quantified metrics.

The full 14-variable disclosed behavioral profile by disclosure archetype is in [Table 22](#) below; the 6 headline variables are shown as [Table 15](#) in Chapter 4.

Table 22: Full disclosed behavioral profile by disclosure archetype (14 variables).

Behavioral feature	Vocal Substantives (n=124)	Governance-Led Disclosers (n=142)	Defensive Disclosers (n=191)	No AI (n=53)
Disclosed activities per firm (Median)	96	18	8	1
Disclosed activities per firm (Mean)	170.7	23.5	15.1	3.6
Customer-facing activity share (%)	52.1%	23.5%	23.9%	21.7%
Internal process activity share (%)	31.1%	63.6%	62.3%	71.2%
Technology source: Proprietary / Own (%)	72.5%	47.2%	38.9%	33.8%
Technology source: Third-party commercial (%)	9.0%	14.8%	17.7%	14.3%
Technology source: Unspecified provenance (%)	18.5%	38.0%	43.4%	51.9%
Firms naming an own product or brand (%)	98.4%	66.2%	43.5%	15.1%
Firms naming an external provider or partner (%)	89.5%	47.2%	31.4%	15.1%
Firms reporting compute / data center capex (%)	89.5%	51.4%	39.3%	7.5%
Firms reporting quantified outcome metrics (%)	93.5%	62.7%	49.7%	24.5%
Firms naming corporate clients (%)	48.4%	9.2%	5.8%	1.9%
Maximum operational stage: Scaled (%)	94.4%	64.1%	45.5%	17.0%
Maximum operational stage: Deployed (%)	5.6%	31.7%	39.3%	20.8%
Maximum operational stage: Pilot / Exploration (%)	0.0%	1.4%	5.2%	11.3%
Maximum operational stage: No activity reported (%)	0.0%	0.7%	6.3%	49.1%

The full prefilter specification, semantic anchors, extraction prompts, and throughput diagnostics are documented in docs/analytics/10_pipeline.md. Corpus rows below are refreshed from the current processed data; downstream rows report the current available inference outputs.

Table 23: Complete granular inference funnel by document form.

Stage	Operational Filter	Unit	10-K	10-Q	DEF 14A	8-K	Calls	Total
A. Corpus								
Documents	Processed corpus	Documents	2,898	8,168	2,880	35,880	7,947	57,773
Paragraphs	Segmentation / turns	Instances	1,555,923	1,774,755	3,992,212	1,693,503	554,281	9,570,674
Scorable paragraphs	Rule: > 3 char & alnum	Instances	1,432,176	1,623,932	3,525,195	1,595,325	553,942	8,730,570
Unique scorable texts	BLAKE2b content hash	Unique texts	861,631	862,147	2,421,944	421,982	483,235	4,979,487
B. Prefilter								
Positive unique texts	GBDT ($\tau = 0.17, 39$ signals)	Unique texts	11,294	2,679	6,344	380	9,584	30,281
Share of scorable texts	Positive rate	%	1.3%	0.3%	0.3%	0.1%	2.0%	0.6%
Positive instances	Exact hash join	Instances	12,487	3,928	6,536	405	9,593	32,952
C. Semantic Frames								
Classified unique texts	Candidate set	Unique texts	11,294	2,679	6,344	380	9,584	30,281

Stage	Operational Filter	Unit	10-K	10-Q	DEF 14A	8-K	Calls	Total
Texts with ≥ 1 frame	LLM judge (Qwen 3.7)	Unique texts	10,532	2,587	5,216	232	7,902	26,469
LLM acceptance rate	Acceptance share	%	93.3%	96.6%	82.2%	61.1%	82.5%	87.4%
Unique semantic frames	Frame extraction	Frames	16,245	3,409	7,168	295	16,249	43,366
Document instances	Exact hash join	Instances	17,758	4,590	7,284	313	16,270	46,215
Documents with ≥ 1 frame	Document join	Documents	1,681	1,431	1,090	206	2,709	7,117
Share of form documents	Document share	%	58.5%	17.7%	41.2%	0.6%	34.1%	12.7%
D. Disclosed Activities								
Texts with firm conduct	Rule: subject = firm	Unique texts	5,927	1,829	2,923	134	6,803	17,420
Share of framed texts	Behaviour share	%	56.3%	70.7%	56.0%	57.8%	86.1%	65.8%
Unique activities	LLM pass 2 (firm ID)	Activities	8,556	2,408	2,670	163	14,020	27,566
Activities per text	Activity density	Ratio	1.44	1.32	0.91	1.22	2.06	1.58
Document instances	Exact hash join	Instances	9,224	3,238	2,708	165	14,033	29,368
Documents with ≥ 1 activity	Document join	Documents	1,298	961	586	88	2,444	5,377
Firms with ≥ 1 activity	Ticker join	Firms	409	159	269	55	383	471 (of 510)

Notes: Complete granular funnel executed on duckdb/thesis.duckdb. Unique texts across columns do not sum to total because identical text paragraphs appear across multiple document forms.

The full distribution of disclosed activities across all six dimensions (27,566 activities, [Table 24](#) below) underlies the activity-evolution chart in Chapter 3 (“Aggregate behavioral landscape”).

Table 24: Full distribution of disclosed activities across all six dimensions.

Action	%	Maturity stage	%	Target audience	%
deploy	62.7%	deployed	58.2%	customers	48.5%
invest infrastructure	7.9%	scaled	14.0%	internal process	25.5%
scale	7.4%	piloting	10.3%	unspecified	13.7%
pilot or explore	4.0%	exploring	9.2%	employees	8.6%
integrate	3.5%	unspecified	8.2%	partners or suppliers	2.5%
develop	3.5%			developers	1.3%
measure outcome	3.4%				
partner	3.0%				
hire or train	2.1%				
buy or license	1.3%				
govern or control	1.2%				
restrict	0.0%				

AI source	%	Evidence strength	%	Business function	%
own	73.0%	generic	53.4%	unspecified	31.9%
unspecified	15.6%	named product or process	29.3%	operations and supply chain	16.5%
third party	7.9%	metric	12.9%	product feature	7.6%
acquired	1.5%	vendor	4.4%	it and cybersecurity	7.0%
mixed	1.0%			other	6.9%
co developed	0.9%			customer service	4.6%
open source	0.1%			software development	4.6%
				marketing and sales	4.1%
				content and media	2.8%
				fraud and risk	2.6%
				data and analytics	2.4%
				research and product development	2.2%
				financial services operations	1.8%
				hr and talent	1.8%
				healthcare delivery	1.1%
				ai compute and models	1.0%
				finance legal and admin	0.8%
				governance	0.5%

Notes: Source: data/processed/clusters/activity_profiles.json. “Business function” has no single dominant category comparable to customer_service for customer-success work specifically. Customer success (retention/expansion) is not coded as a separate family from customer support in the current taxonomy.

APPENDIX D: VALIDATION

The human coding protocol for the pending validation round (300 frame-labeled paragraphs / 434 frames, 300 prefilter-stratified paragraphs, 120 activity-labeled paragraphs / 267 activities, coded in the ui-validator/ tool), per-field agreement statistics once collected, confusion matrices for the promotional and temporal fields, an error taxonomy distinguishing prefilter false positives from judge misclassification, and a sensitivity analysis for collapsing or dropping fields with weak human agreement are reported here as they become available; at the time of writing, only LLM-to-LLM agreement is measured (κ 0.81–0.87, docs/analytics/00_funnel_del_corpus.md).

APPENDIX E: STATISTICAL ROBUSTNESS

Table 25: Full M0-M3 model fit, all five outcomes.

Outcome	R2 M0	R2 M1	R2 M2	R2 M3	Adj.R2 M0	Adj.R2 M1	Adj.R2 M2	Adj.R2 M3
Beta	0.039	0.095	0.125	0.1	-0.098	-0.035	-0.008	-0.033
Idiosyncratic volatility	0.182	0.211	0.225	0.213	0.065	0.098	0.107	0.098
Price-to-sales	0.233	0.235	0.25	0.236	0.125	0.126	0.136	0.124
R&D/sales	0.453	0.479	0.51	0.496	0.409	0.435	0.462	0.451
Revenue growth (t+1)	0.021	0.067	0.116	0.07	-0.122	-0.071	-0.023	-0.071

Notes: Beta and growth start with negative adjusted R^2 at M0 because sector-by-year fixed effects consume many degrees of freedom relative to firms per cell, a penalty the adjustment applies even though raw R^2 stays positive throughout.

Table 26: Standardized coefficients, M2 content block (bold = $p < 0.05$).

Content dimension	Beta	Idiosyncratic volatility	Price-to-sales	R&D/sales	Revenue growth (t+1)
Volume	+0.89 (p=0.02)	+0.51 (p=0.19)	+0.25 (p=0.40)	+0.72 (p<0.01)	-0.07 (p=0.75)
Realized	-0.08 (p=0.82)	-0.31 (p=0.39)	-0.20 (p=0.46)	+0.24 (p=0.42)	+0.09 (p=0.70)
Deployment	-0.55 (p<0.01)	-0.04 (p=0.79)	-0.09 (p=0.57)	-0.37 (p=0.03)	-0.14 (p=0.34)
Capability	-0.13 (p<0.01)	-0.07 (p=0.10)	-0.04 (p=0.55)	-0.11 (p=0.04)	-0.24 (p<0.01)
Risk	-0.20 (p=0.04)	-0.13 (p=0.11)	+0.01 (p=0.92)	-0.26 (p<0.01)	+0.06 (p=0.48)
Governance	-0.13 (p=0.03)	-0.05 (p=0.29)	-0.11 (p=0.04)	-0.11 (p=0.04)	-0.03 (p=0.39)
Promotional	+0.07 (p=0.43)	-0.02 (p=0.81)	-0.09 (p=0.20)	+0.10 (p=0.13)	-0.16 (p=0.02)
Specificity	+0.21 (p=0.11)	+0.24 (p=0.04)	+0.25 (p=0.13)	-0.17 (p=0.12)	+0.64 (p<0.01)

AI-washing index, full specification. $\text{Substance}_{it} = \log(1 + n_{\text{activities},it}) \times \text{Grounding}_{it}$, where Grounding_{it} averages, with empirical-Bayes shrinkage per component, five concreteness markers over that firm-year's disclosed activities (named function, deployed/scaled stage, named product or process, quantified outcome, named provider). $W_{it} = \text{pctrank}(\text{Disclosure}_{it}) - \text{pctrank}(\text{Substance}_{it})$, where Disclosure_{it} is AI frames per 1,000 words and both percentile ranks are computed within the analytical sample (firm-years with $\text{Disclosure}_{it} > 0$: 1,449 of 2,475 firm-years, 474 of 510 firms pooled 2021–2025). No regression, no fixed effects, no fitted residual: it is a rank-based gap between two independently extracted quantities.

Percentile rank rather than z-score standardization: with z-scores, Disclosure_{it} 's distribution has a far heavier right tail than Substance_{it} 's (maximum z of 13.7 against 3.7), which put firms with the *most* grounded activity in absolute terms (Microsoft, NVIDIA) in the washing tail whenever their disclosure happened to be an even bigger outlier than their already-large substance — Spearman between the z-score and percentile-rank versions of W is 0.988, but the specific firm-years in the top 5% differ,

and only the percentile version keeps the tail composed of firms with no grounded activity behind unusually heavy disclosure.

Substance functional form. An additive form, $z(\log(1 + n_{\text{activities}})) + z(\text{Grounding})$, and a grounded-count form, $\log(1 + n_{\text{activities}} \times \text{Grounding})$, were both compared against the multiplicative form used here. Substance under the multiplicative and grounded-count forms correlates at Spearman = 0.993 ($n = 1,449$), and the resulting W at Spearman = 0.988, with 66 of 73 top-5%-tail firm-years shared between the two (Jaccard = 0.825). The multiplicative form is used because it does not reward raw activity count over grounding quality when the two trade off directly against each other (a firm with many but poorly grounded activities does not automatically outscore one with fewer but well-grounded activities, at typical values in this sample), while the grounded-count form's arithmetic can favor the larger, less-grounded firm in exactly that comparison.

Table 27: AI-washing index validation summary (firm-year panel, $n = 1,449$ firm-years disclosing AI).

Check	Sample	Spearman	p
Construct consistency vs. Grounding	$n=1074$	-0.588	<0.001
Persistence (W_t vs. W_{t+1})	$n=929$ pairs	0.602	<0.001
Split-half (years split per firm)	$n=406$ firms	0.514	<0.001
Economic coherence: R&D intensity	$n=718$	-0.089	0.017
Economic coherence: Log market cap	$n=1253$	0.015	0.600
Economic coherence: Beta	$n=1372$	-0.088	0.001
Economic coherence: Idiosyncratic volatility	$n=1372$	-0.099	<0.001
Economic coherence: Price-to-sales	$n=1228$	-0.032	0.255

Notes: Construct consistency against Grounding is restricted to firm-years with at least one disclosed activity ($n = 1,074$); Grounding is a direct component of Substance, so this is not an independent validity check, only confirmation the score moves as its construction implies. The five economic-coherence rows test whether W tracks R&D intensity (expected negative) without also tracking size, valuation, or risk (expected flat) — the pattern that survives is a small, correctly signed R&D relationship and no material size or risk relationship.

Table 28: Firm-years in the top and bottom 5% of the AI-washing index (panel specification), with disclosure, activity, and grounding components.

Ticker	Year	AI frames/1k words	Activities	Grounding	W	Disclosure archetype (Chapter 4)
Q	2025	0.38	0	0.32	0.83	Vocal Substantives
BLK	2024	0.22	0	0.32	0.78	Governance-Led Disclosers
SWKS	2025	0.11	0	0.32	0.67	Defensive Disclosers
WRB	2025	0.09	0	0.32	0.63	Defensive Disclosers
IFF	2024	0.08	0	0.32	0.62	Defensive Disclosers
ROK	2024	0.08	0	0.32	0.61	Governance-Led Disclosers
TER	2025	0.17	1	0.15	0.6	Vocal Substantives
LRCX	2025	0.08	0	0.32	0.6	Defensive Disclosers
IR	2025	0.14	1	0.15	0.57	Defensive Disclosers
LRCX	2024	0.07	0	0.32	0.56	Defensive Disclosers
VRSN	2025	0.07	0	0.32	0.55	Defensive Disclosers
KIM	2024	0.07	0	0.32	0.55	Defensive Disclosers
PM	2024	0.06	0	0.32	0.54	Governance-Led Disclosers
ANSS	2021	0.06	0	0.32	0.54	Vocal Substantives
IT	2024	0.12	4	0.1	0.53	Defensive Disclosers
BBY	2024	0.06	0	0.32	0.53	Defensive Disclosers

Ticker	Year	AI frames/1k words	Activities	Grounding	W	Disclosure archetype (Chapter 4)
KLAC	2023	0.06	0	0.32	0.52	Defensive Disclosers
CLX	2024	0.06	0	0.32	0.52	Defensive Disclosers
CCI	2024	0.06	0	0.32	0.51	Defensive Disclosers
GNRC	2025	0.18	2	0.17	0.5	Defensive Disclosers
DUK	2025	0.05	0	0.32	0.49	Governance-Led Disclosers
WDC	2025	0.14	1	0.26	0.48	Governance-Led Disclosers
LMT	2024	0.1	3	0.13	0.48	Governance-Led Disclosers
DGX	2023	0.08	3	0.07	0.48	Vocal Substantives
IP	2025	0.05	0	0.32	0.48	Defensive Disclosers
IFF	2025	0.13	1	0.26	0.48	Defensive Disclosers
IT	2025	0.17	3	0.17	0.47	Defensive Disclosers
CMCSA	2025	0.05	0	0.32	0.47	Defensive Disclosers
WM	2025	0.05	0	0.32	0.47	Defensive Disclosers
REG	2025	0.05	0	0.32	0.46	Defensive Disclosers
GWW	2025	0.08	4	0.1	0.45	Defensive Disclosers
AMP	2025	0.05	0	0.32	0.45	Governance-Led Disclosers
KLAC	2024	0.26	2	0.24	0.45	Defensive Disclosers
DG	2025	0.05	0	0.32	0.45	Governance-Led Disclosers
CLX	2025	0.05	0	0.32	0.45	Defensive Disclosers
ROP	2025	0.13	2	0.17	0.44	Vocal Substantives
FOX	2025	0.09	1	0.26	0.44	Defensive Disclosers
CTAS	2025	0.04	0	0.32	0.44	Defensive Disclosers
HLT	2025	0.04	0	0.32	0.44	Defensive Disclosers
FOX	2024	0.09	1	0.26	0.43	Defensive Disclosers
DIS	2024	0.04	0	0.32	0.43	Defensive Disclosers
GS	2025	0.07	3	0.07	0.43	Defensive Disclosers
LNC	2024	0.04	0	0.32	0.43	Defensive Disclosers
GM	2024	0.04	0	0.32	0.43	Vocal Substantives
ROP	2024	0.1	1	0.26	0.42	Vocal Substantives
PM	2023	0.04	0	0.32	0.42	Governance-Led Disclosers
DXCM	2023	0.04	0	0.32	0.42	Vocal Substantives
TER	2024	0.12	1	0.35	0.42	Vocal Substantives
DDOG	2021	0.04	0	0.32	0.42	Vocal Substantives
BBY	2025	0.21	4	0.18	0.42	Defensive Disclosers
CHRW	2024	0.23	3	0.23	0.41	Vocal Substantives
VMRK	2025	0.07	2	0.1	0.41	Defensive Disclosers
HLT	2024	0.04	0	0.32	0.41	Defensive Disclosers
BAX	2025	0.16	9	0.11	0.41	Defensive Disclosers
DHR	2024	0.07	1	0.15	0.41	Defensive Disclosers
BSX	2025	0.04	0	0.32	0.41	Defensive Disclosers
LRCX	2022	0.04	0	0.32	0.41	Defensive Disclosers
GILD	2025	0.04	0	0.32	0.4	Governance-Led Disclosers
DIS	2025	0.04	0	0.32	0.4	Defensive Disclosers
DD	2024	0.04	0	0.32	0.4	Governance-Led Disclosers
VLO	2025	0.04	0	0.32	0.4	Governance-Led Disclosers
DHR	2025	0.1	2	0.17	0.4	Defensive Disclosers
SWKS	2024	0.07	1	0.26	0.39	Defensive Disclosers
CTAS	2024	0.04	0	0.32	0.39	Defensive Disclosers
NEE	2025	0.04	0	0.32	0.39	Governance-Led Disclosers
SWKS	2023	0.04	0	0.32	0.39	Defensive Disclosers
D	2025	0.04	0	0.32	0.39	Governance-Led Disclosers
KIM	2025	0.08	1	0.26	0.39	Defensive Disclosers

Ticker	Year	AI frames/1k words	Activities	Grounding	W	Disclosure archetype (Chapter 4)
MCD	2025	0.06	4	0.06	0.39	Defensive Disclosers
TDY	2024	0.1	2	0.17	0.39	Defensive Disclosers
A	2025	0.1	4	0.14	0.39	Defensive Disclosers
AVY	2025	0.04	0	0.32	0.39	Governance-Led Disclosers
GPC	2025	0.04	0	0.32	0.39	Defensive Disclosers
LIN	2025	0.01	2	0.44	-0.37	Governance-Led Disclosers
CPRT	2023	0.01	1	0.46	-0.37	Vocal Substantives
CPRT	2021	0.01	1	0.46	-0.38	Vocal Substantives
NLSN	2021	0.02	3	0.43	-0.38	Governance-Led Disclosers
IRM	2021	0.01	1	0.36	-0.38	Vocal Substantives
CEG	2022	0.01	1	0.46	-0.38	Defensive Disclosers
EW	2022	0.02	2	0.51	-0.38	Defensive Disclosers
AIG	2024	0.01	2	0.31	-0.38	Governance-Led Disclosers
CAH	2024	0.03	5	0.49	-0.39	Governance-Led Disclosers
CPRT	2022	0.01	1	0.46	-0.39	Vocal Substantives
WFC	2024	0.01	2	0.5	-0.39	Vocal Substantives
CVS	2021	0.01	3	0.38	-0.39	Governance-Led Disclosers
FMC	2023	0.01	1	0.46	-0.39	Governance-Led Disclosers
CVS	2023	0.03	5	0.46	-0.39	Governance-Led Disclosers
TSLA	2021	0.01	4	0.34	-0.39	Vocal Substantives
CFG	2022	0.01	2	0.38	-0.4	Governance-Led Disclosers
BNY	2024	0.01	3	0.33	-0.4	Governance-Led Disclosers
FDX	2022	0	1	0.36	-0.4	Vocal Substantives
CL	2023	0.02	4	0.39	-0.4	Governance-Led Disclosers
FBHS	2023	0.01	2	0.38	-0.4	Governance-Led Disclosers
ILMN	2022	0	1	0.36	-0.41	Defensive Disclosers
NEM	2025	0.01	3	0.38	-0.41	Governance-Led Disclosers
SRE	2022	0	1	0.36	-0.41	Defensive Disclosers
ZTS	2025	0.03	6	0.47	-0.41	Vocal Substantives
DOW	2024	0.02	2	0.58	-0.41	Governance-Led Disclosers
ZS	2022	0.01	1	0.45	-0.41	Vocal Substantives
SRE	2021	0	1	0.36	-0.41	Defensive Disclosers
EW	2021	0.01	2	0.5	-0.42	Defensive Disclosers
CRL	2021	0.01	1	0.46	-0.42	Defensive Disclosers
FISV	2021	0.01	1	0.46	-0.42	Vocal Substantives
WELL	2022	0	1	0.36	-0.42	Defensive Disclosers
TRV	2022	0.01	2	0.31	-0.42	Defensive Disclosers
WELL	2023	0	1	0.36	-0.42	Defensive Disclosers
HOLX	2021	0.03	5	0.62	-0.42	Vocal Substantives
ZS	2021	0.01	1	0.45	-0.42	Vocal Substantives
FCX	2021	0	1	0.36	-0.42	Governance-Led Disclosers
ICE	2021	0.03	6	0.56	-0.42	Vocal Substantives
CFG	2023	0	1	0.36	-0.43	Governance-Led Disclosers
SYF	2021	0	1	0.36	-0.43	Governance-Led Disclosers
CSCO	2021	0.02	5	0.46	-0.43	Vocal Substantives
FLT	2023	0.01	3	0.38	-0.43	Governance-Led Disclosers
CTXS	2021	0.01	2	0.51	-0.43	Vocal Substantives
CRL	2024	0.01	1	0.46	-0.43	Defensive Disclosers
EBAY	2021	0.01	2	0.31	-0.43	Vocal Substantives
CFG	2021	0	1	0.36	-0.43	Governance-Led Disclosers
C	2021	0	1	0.36	-0.43	Governance-Led Disclosers
AES	2023	0	1	0.38	-0.44	Vocal Substantives

Ticker	Year	AI frames/1k words	Activities	Grounding	W	Disclosure archetype (Chapter 4)
UHS	2024	0	1	0.37	-0.44	Governance-Led Disclosers
CRL	2023	0.01	1	0.46	-0.44	Defensive Disclosers
BIIB	2022	0.01	2	0.45	-0.45	Governance-Led Disclosers
ALK	2021	0.01	1	0.46	-0.46	Governance-Led Disclosers
MDT	2021	0.02	6	0.39	-0.47	Vocal Substantives
CVS	2022	0.01	3	0.43	-0.47	Governance-Led Disclosers
CARR	2022	0.01	2	0.59	-0.47	Vocal Substantives
WYNN	2021	0.01	1	0.58	-0.48	Governance-Led Disclosers
MMM	2022	0.01	2	0.51	-0.49	Governance-Led Disclosers
HON	2023	0	1	0.46	-0.49	Governance-Led Disclosers
RF	2025	0.03	18	0.56	-0.49	Governance-Led Disclosers
AWK	2024	0.01	6	0.19	-0.5	Vocal Substantives
ICE	2022	0.02	7	0.54	-0.5	Vocal Substantives
TFC	2022	0	1	0.46	-0.5	Governance-Led Disclosers
GE	2023	0	1	0.46	-0.51	Defensive Disclosers
SYF	2022	0	1	0.48	-0.52	Governance-Led Disclosers
KHC	2022	0.01	2	0.38	-0.52	Governance-Led Disclosers
CARR	2023	0.01	3	0.49	-0.52	Vocal Substantives
GNRC	2021	0.02	5	0.52	-0.53	Defensive Disclosers
EIX	2025	0.01	3	0.38	-0.53	Governance-Led Disclosers
CFG	2024	0	2	0.38	-0.55	Governance-Led Disclosers
O	2021	0	2	0.44	-0.6	Governance-Led Disclosers
AEP	2025	0	2	0.45	-0.6	Governance-Led Disclosers
GE	2021	0.02	7	0.54	-0.61	Defensive Disclosers
O	2023	0.01	4	0.46	-0.62	Governance-Led Disclosers
CI	2021	0.01	4	0.46	-0.65	Governance-Led Disclosers

Notes: Positive W is the washing tail (disclosure ranks higher than substance); negative W is the quiet-substance tail (substance ranks higher than disclosure). Both tails together cover 146 of the 1,449 scoreable firm-years (top and bottom 5%).

Alternative firm-year aggregation. Results are unchanged when firm-year intensity is constructed as an aggregate-numerator-over-aggregate-denominator ratio rather than the median document rate: the two rollups are numerically identical across all seven content dimensions and 2,475 firm-years (max absolute difference below $1e-9$), because a firm-year's AI content comes from a single dominant filing often enough that the rollup rule carries no weight in practice.

APPENDIX F: VARIABLE CONSTRUCTION

Table 29: XBRL concept coverage.

Financial concept	Coverage of the firm-year panel
Revenue	98%
Net margin	98%
Capex	87%
SG&A	79%
R&D expense	45% (1,303 of 2,898 firm-years; outcome only)

Fiscal-period alignment rules (period-end date matched to the nearest prior filing date; the [340, 380]-day validity window for year-over-year growth), the market-model beta and idiosyncratic-

volatility construction (252 trading days pre-filing, one-factor market model), and the ROIC-minus-WACC construction (geometric equity risk premium of 6.48%, not arithmetic) are documented here.

8.0.1 R&D COVERAGE AND INTERPRETATION

R&D expense is constructed from four comparable current-expense concepts: `us-gaap:ResearchAndDevelopmentExpense`, `us-gaap:ResearchAndDevelopmentExpenseExcludingAcquiredInProcessCost`, `us-gaap:ResearchAndDevelopmentExpenseSoftwareExcludingAcquiredInProcessCost`, and `ifrs-full:ResearchAndDevelopmentExpense`. The latter two recover software reporters and IFRS filers that the narrower US-GAAP mapping missed; they add 34 firm-years to the panel. The resulting measure is available for 1,303 of 2,898 firm-years (45.0%).

This is not evidence that the remaining firms spend zero on R&D. In a SIC-based diagnostic grouping, separately reported R&D is common among manufacturing and software firms (81.4% coverage) and almost absent in financial services and real estate (1.1%). The missingness is therefore primarily an economic and disclosure-selection feature of the accounting data, rather than a generic failure of the XBRL collection process. The raw stored facts contain no company-specific extension namespace, so this pipeline cannot recover issuer-defined R&D extensions from that input.

For that reason, R&D-to-sales is not imputed and is excluded from the common set of financial controls. When it is an outcome, estimates are interpreted for the non-random subsample of firms that separately disclose comparable current R&D; each specification reports its effective sample size. Tags for tax credits, capitalized development, acquired in-process R&D, and industry-specific exploration costs are deliberately not pooled into this measure because they do not represent the same current R&D expense.

8.0.2 MARKET BETA AND ESTIMATION-WINDOW DIAGNOSTICS

For each firm-year, market beta is estimated from a one-factor market model over the 252 trading days strictly before the actual filing date of the 10-K. Let $R_{i,t}$ denote the daily return on the firm's adjusted close, $R_{f,t}$ the daily risk-free rate, and $MKT - RF_t$ the daily market excess return from the Ken French Fama–French three-factor file (`ff3_daily.parquet`). The estimated regression is

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(MKT - RF_t) + \varepsilon_{i,t}.$$

`build_market_factors.py` identifies the first trading day on or after the filing date and takes the preceding 252 price observations. It then retains only dates for which the firm return, $MKT - RF_t$, and $R_{f,t}$ are all available, and estimates the intercept and slope by OLS. The code requires at least 120 valid daily observations (`BETA_MIN_OBS = 120`); otherwise beta is recorded as missing. The idiosyncratic-volatility measure uses the standard deviation of residuals from the same fitted model, annualized by $\sqrt{252}$.

The resulting `firm_year_market_factors.parquet` panel contains 2,762 firm-years for 477 tickers in 2021–2026. Beta is available for 2,739 observations (99.2%). The estimation-window diagnostic has a mean of 249.8 return days and a median of 252; 98.0% of estimates use at least 240 trading days. The remaining 23 firm-years have fewer than 120 valid observations and are retained with beta missing rather than receiving a short-window estimate. Among valid estimates, beta has mean 0.91, median 0.88, standard deviation 0.47, interquartile range 0.59–1.18, and range $[-0.31, 2.96]$.

Daily prices are collected from 2020-01-01, so early-2021 filings have a prior-year trading history available for the 252-day window. The sampling frame is built around S&P 500 membership frozen at 2021-12-31, supplemented by the manual core; 29 of the 517 firms in the current universe are explicitly marked delisted. Price collection first uses Yahoo Finance and then falls back to Stooq and Tiingo when a delisted ticker is unavailable. Freezing membership and retaining these firms reduces

survivorship bias relative to a current-constituent sample, although historical price coverage still depends on the availability and adjustment quality of those fallback sources.

APPENDIX G: SEC / DEEPSEEK SECONDARY SPECIFICATIONS

Secondary SEC event-study specifications (alternative window widths, weighted variants, cells excluding mixed pre/post filings) and alternative event-window specifications for both shocks are reported here in full, consistent with the one-sentence summary in Chapter 6: no discontinuity separable from prior trends was identified for DeepSeek, and the analysis is not extended to outcome variables (competitive response, generative-AI capex) that a valid identification would be needed to interpret.

Table 30: DeepSeek event study (2025Q1 cutoff, 5,287 firm-quarters, 484 firms): pre-trend test by outcome.

Outcome	Pre-trend test p-value	Fails at 5%?
Promotional	0.017	Yes
Specificity	<0.001	Yes
Risk	<0.001	Yes
Hypothetical framing	0.009	Yes
Total AI frames	<0.001	Yes

Notes: All five outcomes fail the pre-trend test, so no discontinuity around the January 2025 shock can be separated from a pre-existing trend for any of them.